

Financial Supermarkets?

Cross-Selling in Retail Banking

Shengmao Cao

Lulu Wang *

June 20, 2026

Abstract

We study how consumers' tendency to hold multiple financial products from the same bank shapes competition and policy in U.S. retail banking. We document pervasive cross-holding across deposits, credit cards, auto loans, mortgages, and brokerage services — a pattern that spans all consumer segments. Combining an original consumer survey on search and switching with an equilibrium model of bank choice under incomplete consideration, we identify three forces behind cross-holding: correlated preferences across banking products, consideration spillovers from banks' cross-selling efforts, and utility complementarities between same-bank products. Consideration spillovers and utility complementarities together account for a substantial share of bank franchise value, with reward credit cards functioning as a loss leader whose relationship value far exceeds its standalone margin. Accounting for these synergies qualitatively changes standard policy conclusions. A narrow-bank entrant into the deposit market competes at a steep disadvantage, as incumbents are shielded by their credit card customer bases. Interchange-fee regulation that depresses credit card demand spills over into deposit competition. And a horizontal merger that appears anticompetitive on a deposit-HHI screen becomes pro-competitive once the merged bank deploys its credit card as a loss leader for a larger depositor base.

*Cao: Northwestern Kellogg (email: shengmao.cao@kellogg.northwestern.edu). Wang: Northwestern Kellogg (email: lulu.wang@kellogg.northwestern.edu). We thank participants at the Midwest IO Fest and IIOC for their helpful feedback. Marziyeh Aghaei, Lucia Bertoletti, Keigo Makino, Rodrigo Surraco, and Tongtong Yao provided excellent research assistance. All errors are our own.

I Introduction

Cross-selling — the promotion of additional financial products to existing customers — is a cornerstone of modern retail banking. The 1998 merger of Travelers Group and Citicorp sought to create a “financial supermarket”: a one-stop shop for all consumer financial needs. By 2014, Chase’s retail banking division was generating approximately \$4 billion in annual revenue from cross-selling, with roughly half arising from synergies between its credit card and consumer banking operations (JPM, 2015). Cross-selling drew widespread criticism after the 2016 Wells Fargo fake account scandal, yet it has remained a defining feature of the retail banking landscape.

Despite this practical significance, academic research and regulatory policy have largely approached each banking product market in isolation. The literature on bank franchise value, for example, focuses primarily on deposit market power (Egan et al., 2017; Drechsler et al., 2021). The literature on credit-card markets focuses on how interest-rate caps and interchange-fee regulation affect consumers’ use of credit (Agarwal et al., 2015; Wang, 2023). Bank merger reviews, following longstanding federal guidelines, similarly assess competitive harm through deposit market concentration alone (U.S. Department of Justice, 1995; U.S. Department of Justice and Federal Trade Commission, 2023). These studies offer valuable insights but share a common blind spot: how regulatory changes or shifts in competitive structure within one product market spill over to others.

In this paper, we examine how consumers build portfolios of financial products and assess the implications of cross-selling for competition and policy in retail banking. First, using three U.S. consumer surveys, we document empirical patterns of cross-holding — covering deposit accounts, credit cards, mortgages, auto loans, and brokerage services — across banks. Second, we develop and estimate a model of how consumers choose deposit accounts and credit cards over time. Third, we use the estimated model to derive new insights on the sources of banks’ franchise value, policy spillovers between deposit and credit-card markets, and the competitive effects of bank mergers.

Our analysis draws on three consumer surveys: the Survey of Consumer Finances (SCF), the MRI-Simmons Ultimate Survey of Americans, and an original survey of U.S. adults conducted on the Prolific platform. The first two are large, nationally representative studies that document consumers’ financial product portfolios across banks. Our Prolific survey complements these with data on consumers’ search processes, account-switching histories, and responses to hypothetical choice scenarios.

Our central fact is that many consumers choose banks, not individual products.

Across the SCF, MRI-Simmons, and Prolific surveys, roughly 35% of consumers with one checking account and one credit card hold both at the same bank — more than five times the 6.4% benchmark implied by independent product choice. Cross-holding is most pronounced at smaller banks: depositors at megabanks like Chase and Bank of America are on average 6 times more likely to hold the bank’s credit card than non-depositors, while at regional banks such as PNC or Citizens this ratio exceeds 60. We document analogous patterns for auto loans, mortgages, and brokerage accounts.

Cross-holding is prevalent across all consumer segments. In particular, we find similar rates among high-credit-score consumers who use credit cards for rewards and among low-credit-score consumers who revolve balances. The relationship banking literature offers a natural explanation for the latter group: deposit accounts generate information about borrower risk, enabling banks to offer better prices or more favorable underwriting to existing customers (Puri et al., 2017; Agarwal et al., 2018; Parlour et al., 2022). This information channel, however, cannot account for cross-holding among the high-credit-score transactors, who face minimal underwriting barriers at any lender. These patterns therefore point to preference heterogeneity and product complementarity, beyond asymmetric information, as the primary drivers of cross-holding.

While these cross-sectional patterns are striking, they alone cannot distinguish between the mechanisms driving cross-holding. Consumers may simply have correlated preferences across products — a consumer drawn to Chase’s branch network and reputation may prefer it for both deposits and credit cards, generating cross-holding without any causal link between the two decisions. Alternatively, cross-holding may reflect causal channels: banks’ cross-selling efforts may steer existing customers toward additional products, and consumers may derive direct utility from consolidating their financial products at a single institution. These mechanisms have fundamentally different implications for competition and regulatory policy (Gentzkow, 2007). Under correlated preferences alone, cross-holding is a form of consumer sorting and carries no strategic implications. When causal mechanisms are operative, the strategic landscape changes: banks find it profitable to use loss leaders to generate demand for complementary products, and consumers’ tendency to bundle insulates incumbents against entry and transmits competitive or regulatory shocks across product markets.

We designed our survey to gather direct evidence on each of these mechanisms. For each respondent’s most recent primary deposit account and credit-card opening, we reconstruct the search process: the prior accounts they held, the banks they considered, their exposure to direct-mail advertising, and the factors they found most important in the decision. We complement this with discrete choice experiments in which respon-

dents indicate how they would react to hypothetical price changes for a focal product and — for cross-holders — how they would respond if their bank discontinued the complementary product. In each scenario, respondents report whether they would remain with their current bank or switch.

The survey responses point to all three mechanisms. Consumers who consider a bank for a deposit account also tend to consider the same bank for a credit card — even when they ultimately chose neither — indicating that consideration is correlated across products. Consumers who already hold a product at a bank are considerably more likely to include that bank in their consideration set for the other product, showing that existing banking relationships shape the search process. A substantial fraction of cross-holders report that they would switch their focal product if their bank discontinued the complementary account, indicating that consumers derive direct utility from holding both products at the same institution.

We then develop an empirical model of consumer choice over deposit accounts and credit cards in which consumers stochastically receive opportunities to switch one product at a time. When a switching opportunity arrives, the consumer forms a consideration set (Goeree, 2008) and selects the utility-maximizing bank. The model embeds all three mechanisms: correlated bank preferences, consideration spillovers from existing product holdings, and direct utility complementarity between same-bank products. Banks set prices in a Nash equilibrium to maximize the present discounted value of future profits, internalizing the cross-product spillovers through which competitive pricing in one market generates demand in the other. We model two consumer types: transactors, who pay their credit card balance in full each month and optimize over the rewards rate, and revolvers, who carry balances and optimize over the APR (Drechsler et al., 2025). We estimate demand via a Monte Carlo EM algorithm (Train, 2008), recovering rich consumer-level heterogeneity in attentiveness, bank preferences, price sensitivity, and taste for cross-holding.

The estimated model allows us to reassess bank franchise value and the role of complementarity in competition and regulation. Credit cards prove far more important for franchise value than accounting profits suggest. Although rewards-card profits account for roughly 16% of accounting profits, eliminating them would cost banks around 26% of their total profits. Even though banks roughly break even on these cards, rewards cards help retain high-income depositors, making them a loss leader for the higher-margin deposit franchise. For large banks, rewards-card margins would be one-third higher were it not for this loss-leader strategy.

Complementarity also limits entry and generates spillovers across markets. A narrow

bank that offers only deposits faces a steep disadvantage: consumers who value cross-holding remain with their multi-product bank even when the entrant offers a higher deposit rate. This helps explain why deposit entry has been rare despite persistently high spreads. The same cross-market linkages shape policy outcomes: interchange-fee regulation that depresses credit-card demand generates significant deposit-market spillovers, because banks rely on credit-card relationships to retain depositors.

Finally, we apply our framework to a horizontal merger between Citi and a super-regional bank to show how complementarity can flip the competitive verdict. Without complementarity, joint ownership increases concentration and raises prices — the standard anticompetitive prediction. Once we account for complementarity, the merger generates cross-selling synergies: each firm’s credit card can now serve as a loss leader for the other’s deposit base, expanding the scope of cross-selling that neither could exploit alone. The combined firm makes credit-card rewards substantially more attractive — using the card as a loss leader across its expanded portfolio — while letting deposit rates drift slightly lower to harvest surplus from its captive base. Rivals respond by raising their own deposit rates — the opposite of what a concentration analysis predicts. The full Nash response is invisible to a standard deposit-HHI screen: the competitive action runs through credit-card pricing and rivals’ defensive responses on deposits, not through deposit concentration alone.

This paper contributes to several strands of literature. We show that cross-selling is a significant source of bank franchise value and market power, a dimension largely underexplored in the industrial organization literature on financial markets. This literature has developed rich models of individual product markets in isolation, including deposit accounts (Egan et al., 2017; Honka et al., 2017; Egan et al., 2025), credit cards (Stango and Zinman, 2016; Nelson, 2018; Cuesta and Sepúlveda, 2021; Galenianos and Gavazza, 2022; Wang, 2023; Drechsler et al., 2025), and wealth management (Hortaçsu and Syverson, 2004; Hastings et al., 2017; Brown et al., 2023). We complement this work by developing an equilibrium model of multi-product bank competition that reveals how cross-selling shapes banks’ strategic incentives and market power.

We provide additional evidence for demand-based theories of economies of scope in retail banking. Most studies on cross-selling in finance have focused on economies of scope in corporate banking (Hellmann et al., 2008; Laux and Walz, 2009; Ivashina and Kovner, 2011; Santikian, 2014; Qi, 2024). In retail banking, Puri et al. (2017); Agarwal et al. (2018); Parlour et al. (2022) argue that information from deposit accounts helps lenders mitigate credit risk, providing one foundation for why banks bundle deposits with consumer credit. We complement this channel by separately identifying the

demand-side drivers of cross-holding — correlated preferences, consideration spillovers, and utility complementarity. Basten and Juelsrud (2023) and Aguirregabiria et al. (2024) document related scope economies in retail banking. We build on this literature by tracing the equilibrium implications of these mechanisms for pricing, entry, and regulatory policy.

Our paper also contributes to the literature on survey-based demand identification (Petrin, 2002; Berry et al., 2004; Grieco et al., 2024). Standard consumer finance datasets do not separately observe the mechanisms behind cross-holding. We design an original survey to fill this gap, collecting data on consumers’ consideration sets, search histories, and stated responses to hypothetical shocks — each targeted at a distinct mechanism and together helping separate persistent preferences from switching inertia. These survey-based moments offer a natural complement to revealed-preference methods, uncovering dimensions of consumer financial behavior that aggregate market shares alone cannot identify.

More broadly, our paper relates to the literature on consumer search and product bundling. On the search side, we build on limited consideration frameworks (Goeree, 2008; Honka et al., 2017; Egan et al., 2025) to show how narrow consideration sets and infrequent reoptimization amplify bank market power and the returns to cross-selling. On the bundling side, we show that consumer preferences for cross-holding deter entry by single-product firms, extending the entry-deterrence logic of Whinston (1989) and Nalebuff (2004) to a demand-driven setting where the barrier arises from consumer preferences rather than deliberate tying by incumbents. The literature on mergers of complements establishes that combining firms with complementary products lowers prices by internalizing positive cross-demand externalities (Economides and Salop, 1992; Song et al., 2017). We show that a merger can *create* complementarity rather than internalizing a pre-existing one: cross-selling synergies generate differential pricing responses across product markets and yield an overall pro-competitive effect on the industry.

The rest of the paper is organized as follows. Section II describes the data. Section III documents stylized facts on cross-holding behavior using three complementary surveys of U.S. consumers. Section IV develops an equilibrium model of consumer choice and bank competition in the deposit account and credit-card markets. Section V describes the estimation strategy and reports the structural estimates. Section VI uses the estimated model to analyze franchise value, entry, interchange-fee regulation, and bank mergers. Section VII concludes.

II Data

We combine three complementary datasets to describe and analyze households’ financial product choices.

Survey of Consumer Finances (SCF) Our first dataset is the 2022 Survey of Consumer Finances (SCF), the latest wave of the triennial survey conducted by the Federal Reserve Board. The SCF is widely regarded as the most comprehensive source on the finances of U.S. households. We use the 2022 Full Public Data Set, which covers the financial positions and demographic characteristics of 4,595 households.

The SCF covers a broad set of financial products, including deposit accounts, credit cards, auto loans, mortgages, and brokerage services. Although bank anonymity has limited its use in prior retail banking research,¹ we exploit a key structural detail: the SCF assigns consistent institution codes across different products within each household. This allows us to identify whether a household holds multiple products at the same institution — for example, a checking account and a mortgage at the same bank. One limitation is that we cannot link cross-holding patterns to specific named banks.

MRI-Simmons USA (MRI) Our second dataset, the MRI-Simmons USA Study (MRI), complements the SCF by identifying the specific financial institutions consumers use. A large-scale, nationally representative survey, MRI provides detailed product-ownership records alongside demographic data. We primarily use the 2021 and 2022 waves (approximately 100,000 consumers) and also draw on the 2020 wave (approximately 40,000 consumers), which captures consumer tenure at their financial institutions.² MRI’s product coverage is narrower than the SCF’s — focused on deposit accounts, credit cards, and brokerage services — but it overcomes the SCF’s anonymity restriction. Nine banks in the data offer both deposit accounts and credit cards: Chase, Bank of America, Wells Fargo, Citi, Capital One, U.S. Bank, PNC, TD Bank, and Citizens. Among these, Chase, Bank of America, and Wells Fargo also offer brokerage services.

Two important limitations remain in both datasets. First, as they are repeated cross-sections, we observe static product portfolios rather than consumers’ search or switching behavior over time. Second, they capture consumers’ cross-holding decisions in equilibrium but not banks’ cross-selling strategies that shape these patterns. Our Prolific survey is designed to address both gaps.

¹The survey anonymizes financial institutions, preventing direct identification of which bank a household uses.

²The 2020 wave is referred to as NHCS.

Original Prolific Survey To overcome these limitations, we fielded an original survey of approximately 955 U.S. consumers on Prolific.³ The survey takes around ten minutes, and each participant receives \$3 upon completion. It mirrors the MRI structure by collecting data on each consumer’s deposit and credit-card holdings, distinguishing between primary and other accounts. As shown in Table 2, our sample tracks aggregate market shares in the deposit and credit-card markets reasonably well, although it modestly overrepresents customers of online-focused banks such as American Express, Capital One, and Discover.

The survey introduces two design features aimed at uncovering the mechanisms behind consumer choices. First, we reconstruct consumers’ search and switching histories. For each respondent’s most recent primary account opening, we elicit the timing of the decision, the banks they considered, the prior accounts they held, exposure to direct-mail advertising, and the factors they found most important in the decision.

Second, we use a stated-preference design to collect information on substitution patterns and complementarities. We ask consumers to identify their ‘ideal’ bank for a specific product if they were to make a choice *de novo* today; differences between this ideal and their current choice indicate inattention or switching costs. We also elicit their ‘second-best’ choice if their ideal bank were not available. Finally, to measure cross-product complementarities, we present respondents with hypothetical shocks: an increase in competitors’ deposit interest rates, a reduction in credit-card rewards, and — for cross-holders — the closure of one account line by their bank. These questions resemble the “discrete choice experiment” design commonly used to estimate the value of non-wage amenities and labor-supply elasticities (Mas and Pallais, 2017, 2019; Caldwell et al., 2025). The resulting switching probabilities provide key empirical moments for estimating the magnitude of demand-side complementarities.

We carefully design the survey to mitigate potential framing and order effects. To prevent consideration-set reports from being contaminated by recall of prior banking relationships, we elicit the consideration set before asking about prior accounts. Similarly, we present the unbundling shock before the price shocks, ensuring that responses are not anchored by the price sensitivity elicited in subsequent questions.⁴

Table 1 compares key characteristics across our data sources. Panel A shows that median incomes are broadly comparable, ranging from \$62k in our Prolific sample to \$71k in the SCF. Our Prolific sample modestly overrepresents college-educated consumers,

³The survey can be accessed at the following link: https://kellogg.qualtrics.com/jfe/form/SV_eXKObVgGMnzCcGa.

⁴Any potential anchoring effects on price sensitivity can be assessed by comparing the responses of cross-holders to those of non-cross-holders, who were not presented with the unbundling shock.

consistent with known selection patterns on online survey platforms. Product ownership rates in Panel B are similar across datasets. Panels C1 and C2 reveal substantial differences in deposit balances and credit-card spending between revolvers (who carry a balance month-to-month) and transactors (who pay in full), a pattern we exploit to calibrate our supply-side model.

Additional Data To complete the supply-side analysis, we collect price data from three additional sources. We combine bank financial filings with Nilson Report payment volume data to estimate average rewards expenses at the bank level. We obtain average credit-card APRs by bank from the CFPB Terms of Credit Card Plans (TCCP) survey. Deposit rates are constructed from the quarterly Y-14 bank holding company filings as the ratio of interest expense on domestic deposits to total domestic deposits.

III Stylized Facts

We document four stylized facts on the prevalence of cross-holding and three channels through which it arises: — correlated preferences, consideration spillovers, and utility complementarity.

We measure cross-holding using two complementary approaches. Our *aggregate cross-holding rate* measures the fraction of consumers who hold a checking account and a second financial product at the same bank. We compute it for consumers with exactly one checking account and exactly one secondary product (e.g., one credit card), so that cross-holding status is unambiguous.⁵ Our *bank-specific conditional share* measures the probability of holding a secondary product at a specific named bank, conditional on holding a checking account there. We compute this measure using the full sample in MRI and our Prolific survey, without restricting to consumers with exactly one account of each type; it cannot be computed from the SCF, where institution identities are anonymous.

Fact 1. *Cross-holding is pervasive across all demographic groups and represents a fivefold increase over a benchmark of independent product choices. Reliance on cross-holding is significantly greater among smaller, regional banks.*

Figure 1 plots the aggregate cross-holding rate for checking accounts paired with credit cards, auto loans, mortgages, and brokerage accounts, using all three datasets

⁵In MRI and our Prolific survey, banks outside a named set are grouped into a residual “Other” category. We exclude consumers for whom both products fall under “Other,” since cross-holding status is then indeterminate. The SCF does not face this issue: although it anonymizes bank names, it assigns each household consistent internal institution codes, allowing us to identify cross-holding status regardless of the institution’s identity.

(SCF, MRI, and Prolific). Approximately 35% of consumers hold their checking account and credit card at the same bank; the analogous rates for auto loans, mortgages, and brokerage accounts are 15, 18, and 8 percent, respectively.⁶ To gauge the magnitude of cross-holding, we construct an independent-choices benchmark using aggregate market shares: if deposit and credit-card choices were independent, only 6.4% of consumers would hold both at the same bank. At 35%, the observed rate is more than five times this benchmark.

The propensity to cross-hold is remarkably stable across demographic groups. Figure 2 shows nearly identical cross-holding rates across income, education, and credit score levels, and between credit-card transactors and revolvers, with the pattern replicated across all three datasets. This uniformity distinguishes our findings from the “information channel” emphasized in prior literature (Puri et al., 2017; Agarwal et al., 2018; Parlour et al., 2022) — the hypothesis that banks leverage deposit data to identify creditworthy borrowers who would face higher underwriting standards elsewhere. Cross-holding is equally prevalent among high-credit-score consumers who face minimal underwriting barriers, and among transactors who optimize over publicly posted reward rates. Together, these patterns suggest the decision to cross-hold reflects preference heterogeneity or product complementarity rather than asymmetric information alone.⁷

Finally, bank-specific conditional shares reveal that smaller banks are disproportionately dependent on cross-holding. Figure 3 plots each bank’s credit-card market share separately among its depositors and non-depositors, using the full MRI sample. We define the *cross-holding ratio* for each bank as the ratio of these two conditional shares. For “megabanks” like Chase and Bank of America, the cross-holding ratio averages 6× (4× for Chase, 8× for Bank of America); for regional banks such as PNC or Citizens, it exceeds 60. For these smaller banks, credit-card adoption is almost entirely driven by the deposit relationship, while megabanks attract a substantial share of standalone credit-card customers. The figure also documents a symmetric pattern: deposit account market shares are similarly elevated among a bank’s credit-card holders. Together, these

⁶The absolute level of the cross-holding rate varies somewhat across datasets because of differences in bank coverage and the share of consumers assigned to unidentified institutions, but the qualitative pattern is consistent.

⁷Cross-holding is negatively associated with financial sophistication. Using the MRI sample, we regress an indicator for cross-holding on consumer characteristics (Appendix Table A.1). Consumers in specialized financial roles — such as CFOs and treasurers — and those who hold stocks or tax-advantaged investments are substantially less likely to cross-hold. This pattern reflects heterogeneity in how consumers weigh the convenience of one-stop banking against the gains from shopping across providers.

results suggest that the “relationship multiplier” is central to customer acquisition at smaller banks.

Fact 2. *Consumers do not continuously optimize their bank choices and switch banks infrequently.*

Figure 4 plots tenure distributions at current banks from our Prolific survey.⁸ The median tenure for a primary deposit account is 8 years; 46% of consumers have banked with the same institution for over a decade, while only 2.3% switched in the past year. The median credit-card tenure is 6 years, with 37% holding the same card for at least a decade. Only 4.5% of consumers opened their primary deposit account and credit card simultaneously, suggesting that consumers typically do not search for or acquire these products together.

The long tenures suggest that banks retain consumers for extended periods once acquired. To gauge whether this stickiness reflects genuine satisfaction or underlying frictions, we asked respondents to identify their “ideal” bank for a deposit account or a credit card if they were to choose *de novo* today. For deposit accounts, 9.9% prefer a bank different from their current provider — more than four times the 2.3% who actually switched accounts in the past year. The same pattern holds for credit cards: 10.7% identify a different “ideal” issuer, compared with only 2.5% who acquired a new primary card in the past year. These gaps between stated preferences and actual choices suggest that infrequent switching partly reflects frictions such as inattention or switching costs (Egan et al., 2025). Our *de novo* elicitation separates genuine preference alignment from consumer inertia, a distinction that observed tenure alone cannot make.

Fact 3. *Consumers are significantly more likely to consider their current bank when searching for a new product.*

Banks actively cross-sell to existing customers through targeted marketing, generating consideration spillovers that help explain the cross-holding patterns documented above. To measure this channel, we use our Prolific survey, in which respondents listed the specific banks they “looked into or reviewed” during their most recent account search. By matching these consideration sets with the consumers’ pre-existing holdings, we quantify the impact of current relationships on the search process.

Consideration sets are narrow: consumers considered an average of only 2.0 banks for deposit accounts and 2.5 for credit cards (Appendix Figure A.3). Table 3 shows clear

⁸We ask each respondent to recall when they opened their primary accounts. Because of imperfect recall and rounding, we observe masses at 5, 10, and 15 years in the reported tenure distributions.

evidence of relationship spillovers: holding one product at a bank raises the probability of considering that bank for a new product by 6 to 11 percentage points. These spillovers expand the overall consideration set rather than crowding out rivals: for example, deposit-holders at a bank offering credit cards considered an average of 2.4 credit-card products, compared to 2.1 among non-holders.

Table 3 also documents correlated consideration across product markets. Conditional on current holdings, consumers who consider a bank for one product are approximately 25 percentage points more likely to consider it for the other, even among those who chose neither product from that bank. This co-consideration likely reflects shared brand awareness — the same advertising or reputation that brings a bank to mind for deposits also brings it to mind for credit cards.

Fact 4. *Forced unbundling triggers switching probabilities comparable to large price shocks, indicating substantial utility complementarities from cross-holding deposit accounts and credit cards.*

Using the stated-preference module of our Prolific survey, we elicit responses to a “discontinuation shock”: we ask cross-holders whether they would switch their primary bank for one product if their bank discontinued the other. We benchmark these responses against reactions to standard price shocks, such as changes in deposit rates, credit-card reward rates, and APRs.

Figure 5 summarizes the results. Approximately 32% of cross-holders would switch their deposit account if their bank discontinued its credit card — comparable to the 40% of transactors and 27% of revolvers who would switch following a 50-basis-point increase in competitors’ deposit rates. For credit cards, 36% would switch if their bank discontinued deposits — a substantial share, though below the 56% of transactors responding to a half-point drop in rewards and 72% of revolvers responding to a 5 percentage-point increase in APRs.

If cross-holding reflected only prior cross-selling or correlated preferences, losing one product should leave demand for the other unchanged. The large fraction of switchers in response to the discontinuation shock — around one-third of cross-holders — provides direct evidence that the convenience of one-stop service is a first-order driver of cross-holding.

IV Model

A dynamic model of consumer choices over deposits and credit cards separates the three mechanisms behind cross-holding: correlated preferences, consideration spillovers,

and utility complementarity. Cross-holding driven solely by correlated preferences carries no strategic implications, while consideration spillovers and utility complementarity both generate loss-leader pricing incentives and entry barriers. Separating the correlational and causal channels is therefore essential for counterfactual policy evaluations.

IV.A Demand

Let i index consumers, j banks, $k \in 1, 2$ product categories ($k = 1$: deposits; $k = 2$: credit cards), and t time. Consumer i 's state is the pair $x_i = (x_{i1}, x_{i2})$ of current bank choices, where $x_{ik} \in \mathcal{J}_k$ denotes the bank currently held for product k .

Consideration and Choices Motivated by Fact 2, we assume consumers are inattentive and do not continuously optimize their choices. Instead, opportunities to reconsider product k arrive stochastically according to a Poisson process with arrival rate λ_{ik} .⁹ Reconsideration opportunities arrive independently across products — consistent with our survey evidence that consumers acquire deposits and credit cards sequentially — so when product k is being reconsidered, the other product's choice $x_{i,-k}$ enters as a state variable in both the consideration and utility functions.

When a switching opportunity for product k arrives at time t , the consumer forms a consideration set $C_{ikt} \subset \mathcal{J}_k$ (Honka et al., 2017). The probability of forming a given consideration set, conditional on state x_i , is the product of individual consideration probabilities:

$$P_{ik}(C_{ikt}|x_i) = \prod_{j \in C_{ikt}} \phi_{ijk}(x_i) \cdot \prod_{j \notin C_{ikt}} (1 - \phi_{ijk}(x_i)), \quad (1)$$

where $\phi_{ijk}(x_i)$ is the probability that bank j enters the consideration set for product k . Consumers automatically include their current bank; for all others, consideration follows a logistic form:

$$\phi_{ijk}(x_i) = \begin{cases} 1 & x_{ik} = j \\ \frac{1}{1 + \exp(-z_{ijk}(x_i))} & x_{ik} \neq j, \end{cases}$$

where the latent index $z_{ijk}(x_i)$ is:

$$z_{ijk}(x_i) = v_{ijk}^c + \chi_k^c \mathbb{1}(x_{i,-k} = j). \quad (2)$$

⁹ λ_{ik} captures the rate at which already-adopted choices are revisited; the timing of *initial* product adoption is taken as given and is not structurally modeled.

The two terms capture, respectively, consumer-specific bank visibility v_{ijk}^c and cross-selling spillovers from existing holdings ($\chi_k^c \mathbb{1}(x_{i,-k} = j)$): holding product $-k$ at bank j raises the probability of considering j for product k (Fact 3). Consistent with our survey evidence, cross-selling expands consideration sets rather than crowding out rivals.

Given a consideration set, consumer i chooses bank $j \in C_{ikt}$ to maximize:

$$u_{ijkt}(x_i) = \underbrace{v_{ijk}^u + \alpha_{ik} p_{jk} + \chi_{ik}^u \mathbb{1}(x_{i,-k} = j)}_{\delta_{ijk}(x_i)} + \varepsilon_{ijkt}, \quad (3)$$

where v_{ijk}^u is the consumer-specific bank-product quality, $\alpha_{ik} p_{jk}$ captures price sensitivity with p_{jk} signed so that higher values are preferred ($\alpha_{ik} > 0$),¹⁰ $\chi_{ik}^u \mathbb{1}(x_{i,-k} = j)$ is the utility complementarity from cross-holding (Fact 4), and ε_{ijkt} is an idiosyncratic match value that follows a Type I Extreme Value distribution.

We allow consumer-level heterogeneity in the bank-preference intercepts v_{ij}^c , v_{ij}^u , price sensitivities α_{ik} , utility complementarities χ_{ik}^u , and arrival rates λ_{ik} , collected in a type vector θ_i . We treat consideration spillovers χ_k^c as homogeneous: the data identify its mean (Fact 3) but not consumer-level variance. We defer the parametric form of the type distribution to Section V.A.

The conditional probability that consumer i chooses bank j from consideration set C_{ik} is:

$$s_{ijk}(x_i, C_{ik}) = \mathbb{1}\{j \in C_{ik}\} \times \frac{\exp(\delta_{ijk}(x_i))}{1 + \sum_{r \in C_{ik} \setminus \{0\}} \exp(\delta_{irk}(x_i))}. \quad (4)$$

Averaging over all possible consideration sets yields the unconditional choice probability $m_{ijk}(x_i)$:

$$m_{ijk}(x_i) = \sum_{C_{ik} \in 2^{\mathcal{J}_k}} P_{ik}(C_{ik}|x_i) s_{ijk}(x_i, C_{ik}) \quad (5)$$

Steady-State Market Shares The choice processes described above generate dynamic cross-product linkages through spillovers across product lines. A bank's success in one market depends on its presence in the other, via both cross-selling effects and utility complementarities. We characterize each consumer's long-run holdings using a continuous-time Markov framework (Arcidiacono et al., 2016), which simplifies the

¹⁰We separately estimate the model for transactors, who pay their balance in full each month and optimize over reward rates, and revolvers, who carry a balance and optimize over APRs. For deposit accounts and transactors, p_{jk} equals the deposit rate and reward rate, respectively. For revolvers, we define $p_{jk} = -\text{APR}$.

high-dimensional transition space by assuming that consumers reconsider one product at a time (Fact 2).

For each product k , let $Q_{ik}(\theta_i)$ denote consumer i 's transition intensity matrix, whose (x, y) th element $q_{ik}^{x \rightarrow y}$ is the rate of moving from state x to state y . Given the arrival rate λ_{ik} and choice probabilities $m_{ijk}(x)$, the transition rates are:

$$q_{ik}^{x \rightarrow y} = \begin{cases} 0, & \text{if } x_{-k} \neq y_{-k}, \\ \lambda_{ik} m_{iy_k k}(x), & \text{if } x_k \neq y_k \text{ and } x_{-k} = y_{-k}, \\ -\lambda_{ik} (1 - m_{ix_k k}(x)), & \text{if } x = y. \end{cases} \quad (6)$$

The first case imposes independent arrivals across products: when a switching opportunity for product k arises, the other product's choice remains unchanged. The remaining elements follow from the choice probabilities and the standard generator-matrix convention.

Under independent Poisson arrivals, the aggregate transition intensity matrix is the sum of product-specific intensities: $Q_i(\theta_i) = Q_{i1}(\theta_i) + Q_{i2}(\theta_i)$. Consumer i 's steady-state holding distribution $\bar{s}_i(\theta_i)$ is the unique row vector satisfying

$$\bar{s}_i(\theta_i) Q_i(\theta_i) = 0, \quad \sum_x \bar{s}_i(x; \theta_i) = 1. \quad (7)$$

Aggregate market shares follow by averaging over the type distribution: $\bar{s}(\theta) = \mathbb{E}_\theta[\bar{s}_i(\theta_i)]$.

Appendix B solves a simple two-bank example with cross-holding complementarity to illustrate how choice probabilities, transition rates, and steady-state market shares are computed under the framework.

IV.B Supply

Two key features shape banks' pricing incentives. First, pricing generates cross-product spillovers: a competitive price on one product attracts customers who may later adopt the other, so that the marginal value of a new customer exceeds the flow profit from any single product. Second, consumer inattention means demand adjusts only slowly to price changes, granting banks temporary market power over their "sleepy" existing customers.

Banks set prices to maximize the present discounted value (PDV) of joint profits across product lines, taking the competitors' prices as given. With consumer heterogeneity, bank j 's profit is the population expectation of the discounted flow payoff:

$$\max_{p_j} \Pi_j(p_j, p_{-j}) = \mathbb{E}_\theta \left[\int_0^\infty e^{-rt} \bar{s}_i(\theta) e^{Q_i(p_j, p_{-j}; \theta)t} h_j(\theta) dt \right]. \quad (8)$$

For each type θ , the integrand is the PDV of flow profits from a consumer starting at steady state $\bar{s}_i(\theta)$ whose holdings evolve at rate $Q_i(p_j, p_{-j}; \theta)$. The flow profit vector is $h_j(\theta) = \sum_{k=1}^2 O_{jk} b_k \pi_{jk}$, where O_{jk} identifies the states in which the consumer holds product k at bank j , π_{jk} is the per-unit margin,¹¹ and b_k is the average account balance calibrated at the transactor/revolver segment level.¹²

Each type's inner integral admits a closed-form solution via the *resolvent* of its generator matrix. At discount rate r , the type-specific resolvent is:¹³

$$\mathcal{R}(p_j, p_{-j}; \theta) \equiv (rI - Q_i(p_j, p_{-j}; \theta))^{-1} = \int_0^\infty e^{-rt} e^{Q_i(p_j, p_{-j}; \theta)t} dt, \quad (9)$$

Applying this identity, the profit function simplifies to:

$$\Pi_j(p_j, p_{-j}) = \mathbb{E}_\theta [\bar{s}_i(\theta) \mathcal{R}(p_j, p_{-j}; \theta) h_j(\theta)]. \quad (10)$$

The resolvent $\mathcal{R}(\cdot; \theta)$ converts the instantaneous transition rates in Q_i into the discounted expected time each consumer spends in each banking state. Equation 10 then averages these type-specific lifetime values across the population.

Our supply-side model is “essentially static”, following Brown et al. (2023). Banks do not set prices period by period based on the current state, but instead set long-run optimal prices, recognizing that demand is slow to respond due to consumer inattention. Lower arrival rates of switching opportunities imply slower adjustments, granting banks greater market power over their “sleepy” existing customers.

First-Order Conditions The resolvent representation yields clean first-order conditions for pricing. Differentiating the profit function (Equation 10) with respect to p_{jk} requires the derivative of each consumer type's resolvent $R_i(\theta)$. Applying the matrix inverse identity $\partial A^{-1} = -A^{-1}(\partial A)A^{-1}$ gives:

¹¹For deposit accounts and credit-card transactors, $\pi_{jk} = r_{jk} - p_{jk}$, where r_{jk} is the investment return on deposits and the interchange fee revenue, respectively. For revolvers, $\pi_{jk} = \text{APR}_{jk} - c_{jk}$, consistent with the demand-side convention $p_{jk} = -\text{APR}_{jk}$, where c_{jk} is the bank's cost of funds from overhead and default risk.

¹²We calibrate b_k as the average of median values across three independent survey sources (SCF, NHCS, and MRI), separately by consumer type. Appendix C provides details of the calibration procedure.

¹³The resolvent is well-defined because each Q_i is a generator matrix with non-positive real eigenvalues and $r > 0$.

$$\frac{\partial \mathcal{R}(\theta)}{\partial p_{jk}} = \mathcal{R}(\theta) \frac{\partial Q_i(\theta)}{\partial p_{jk}} \mathcal{R}(\theta). \quad (11)$$

Since $\partial h_j / \partial p_{jk} = -O_{jk} b_k$, the first-order condition for p_{jk} is:¹⁴

$$\underbrace{\mathbb{E}_\theta \left[\bar{s}_i(\theta) \mathcal{R}(\theta) \frac{\partial Q_i(\theta)}{\partial p_{jk}} \mathcal{R}(\theta) h_j(\theta) \right]}_{\text{Marginal benefit of gained customers}} = \underbrace{\mathbb{E}_\theta [\bar{s}_i(\theta) \mathcal{R}(\theta) O_{jk} b_k]}_{\text{Infra-marginal cost}}. \quad (12)$$

Banks balance the discounted benefit of attracting new customers against the cost of lower margins on the existing customer base. On the benefit side, a more generous price shifts transition rates in the bank's favor, expanding its steady-state customer base across both product lines; because h_j aggregates profits from deposits and credit cards jointly, the bank internalizes that winning a credit-card customer today raises the probability of also earning deposit margins from that same consumer. On the cost side, the bank sacrifices b_k of balance on every existing holder of product k . Because switching opportunities arrive stochastically, a large inframarginal base is slow to respond to less favorable terms, strengthening the bank's incentive to harvest the existing base.

V Estimation and Results

We estimate the model by maximum likelihood, using the survey design to separately identify correlated bank preferences, consideration spillovers, and utility complementarity. All three mechanisms are active and quantitatively important. We then invert the supply-side first-order conditions to recover bank-level marginal costs. The implied margins align with external benchmarks on deposit spreads and credit-card margins.

V.A Parametrization of Consumer Preferences

The bank-product preferences are jointly normal in levels: for each bank j ,

$$(\nu_{ij1}^c, \nu_{ij2}^c, \nu_{ij1}^u, \nu_{ij2}^u) \sim \mathcal{N}(\bar{\nu}_j, \Sigma_\nu),$$

where Σ_ν is shared across banks and allows within-bank correlations across products along both the consideration and utility margins. We treat consideration spillovers χ_k^c as homogeneous across consumers. Because price sensitivities, complementarities, and arrival rates are positive on theoretical grounds, we parameterize them in logs. For each product k , $(\log \alpha_{ik}, \log \chi_{ik}^u)$ is jointly normal with within-product correlation; the

¹⁴We impose separate credit card FOCs for transactors (reward-rate) and revolvers (APR), while pooling the deposit rate FOC across segments.

two product blocks are independent, since the survey randomizes each respondent into hypotheticals for only one product. Arrival rates are jointly normal across products in logs, $(\log \lambda_{i1}, \log \lambda_{i2}) \sim \mathcal{N}(\log \bar{\lambda}, \Sigma_\lambda)$.

V.B Intuition for Identification

The three cross-holding mechanisms are separately identified by distinct features of the survey data.

Correlated consideration and consideration spillovers. Two moments in Fact 3 jointly identify these parameters. The first is co-consideration: consumers tend to include the same bank in their consideration sets for both products, even when they hold neither. This pattern reflects persistent, bank-specific exposure that does not depend on any prior relationship, and pins down the within-bank consideration correlation in Σ_ν . The second is the consideration boost from an existing relationship: after conditioning on this baseline tendency, consumers with a prior account at a bank are disproportionately likely to consider that bank for the other product. This residual elevation — attributable to the existing relationship rather than to brand exposure alone — identifies the consideration spillover χ_k^c .

Utility complementarity. The average utility complementarity $\bar{\chi}_k^u$ is identified by the unbundling shock in Fact 4. We ask cross-holding consumers whether they would switch their deposit account if their bank discontinued its credit card, and vice versa. A consumer who would switch upon losing the complementary product reveals a preference for consolidation that cannot be attributed to brand affinity or prior consideration (Gentzkow, 2007).¹⁵

Correlated preferences. After χ^c and χ^u account for the causal channels, the residual cross-holding tendency — the rate of cross-holding in excess of what the causal channels predict — identifies the within-bank utility correlation $\rho(\nu_1^u, \nu_2^u)$ in Σ_ν . This component captures persistent brand preferences that raise the appeal of both of a bank’s products after the consideration set has been formed.

Preference heterogeneity and arrival rates. Our *de novo* choice question separates persistent preference heterogeneity from consumer inertia. We ask each respondent to name

¹⁵The unbundling question is posed only to current cross-holders, who are selected for high realized χ_i^u . We correct for this by combining the conditional distribution of responses among cross-holders with population cross-holding rates to recover the unconditional distribution of χ_i^u .

their ideal deposit account or credit card if choosing freely today. The would-switch rate — the fraction who name a bank other than their current one — identifies the dispersion of bank preferences: when preferences are strongly dispersed, most consumers already hold their preferred products and few would switch; a high would-switch rate instead signals that inertia keeps many consumers away from their ideal.

Given the estimated preference distribution, the tenure distribution pins down the arrival rates. Its shape is governed by two factors: the reconsideration rate λ_{ik} and the post-wake-up switch probability. With the latter identified by the *de novo* choice questions, the observed shape of the tenure distribution — in particular its heavy right tail — directly identifies the mean and dispersion of λ_{ik} across consumers.

Finally, we identify the distributions of price sensitivity α_{ik} and utility complementarity χ_{ik}^u from stated responses to hypothetical price and unbundling shocks. A respondent who would switch under a small price shock reveals high α_{ik} ; one who would switch upon unbundling reveals high χ_{ik}^u . Because each respondent answers both types of questions for the same product, the joint distribution of responses traces out the within-product correlation between α_{ik} and χ_{ik}^u .

V.C Likelihood Function

We estimate the demand model by maximum likelihood, integrating over the consumer-specific parameters θ_i :

$$\max_{\bar{\theta}, \Sigma_{\theta}} \sum_{i=1}^N \log \int L(X_i | \theta) \phi(\theta; \bar{\theta}, \Sigma_{\theta}) d\theta,$$

where $\phi(\cdot; \bar{\theta}, \Sigma_{\theta})$ is the multivariate normal density specified in Section V.A and $L(X_i | \theta_i)$ is the complete-data likelihood for consumer i . Section V.D describes the Monte Carlo EM algorithm we use to solve this problem.

The complete-data likelihood combines three blocks of information about each Prolific respondent. The static-choices block models the consideration sets and bank choices reported at their most recent product switches. The dynamic block models the timing of those choices, capturing how frequently the consumer reconsiders each product and the duration of each tenure spell. The stated-preference block models their responses to hypothetical price and unbundling shocks.

Static choices (Prolific historical decisions) The Prolific survey reconstructs each respondent’s most recent deposit-account and credit-card openings, recording the bank chosen, the set of banks they “looked into or reviewed” in the search, and their product

holdings at the time of each switch. For each switch k , the likelihood contribution is the joint probability of the elicited consideration set C_k and the chosen bank x_k^* given the pre-switch state $x(t_k^-)$:

$$D_k(\theta_i) = \underbrace{P(C_k \mid x(t_k^-), \theta_i)}_{\text{consideration-set probability}} \cdot \underbrace{P(x_k^* \mid C_k, x(t_k^-), \theta_i)}_{\text{logit choice given consideration}}, \quad (13)$$

where the first term is the product of bank-specific inclusion probabilities and the second is the conditional logit over banks in the consideration set, both specified in Section IV. The static block is the product of the two contributions:

$$L^{\text{stat}}(\theta_i) = D_1(\theta_i) \cdot D_2(\theta_i),$$

Each snapshot conditions on a different pre-switch state, capturing how prior holding shifts the consideration set and the conditional choice (Facts 3 and 4).

Dynamics (Prolific past switching) The static block describes the choice events themselves; the dynamic block adds the timing layer that turns those events into a coherent stochastic process. The Prolific survey records the elapsed time since each switch — for example, t_1 since the deposit switch, t_2 since the credit-card switch, with $t_1 > t_2$ — and we model this timing using the continuous-time Markov framework of Section IV. The dynamic likelihood contribution is

$$L^{\text{dyn}}(\theta_i) = \underbrace{\lambda_{i1}}_{\text{deposit-switch arrival rate}} \times \underbrace{M}_{\text{intervening tenure}} \times \underbrace{\lambda_{i2}}_{\text{credit-switch arrival rate}} \times \underbrace{F}_{\text{final stability}}. \quad (14)$$

where each λ_{ik} is the Poisson rate at which the consumer reconsiders product k , M is the matrix-exponential transition probability over the interval (t_2, t_1) between the two switches, and F is the matrix-exponential probability of remaining in the current state from the most recent switch to the survey date.¹⁶

Stated responses (Prolific hypothetical questions) We model each respondent's stated responses as a revealed ranking of banks under progressively unfavorable conditions. The survey elicits four choices: the ideal bank V , the choice after an unfavorable price change at that bank (W), the choice after losing the complementary product (R), and the

¹⁶Appendix D derives closed-form expressions for M and F , adjustments for first-time adoption, integrals that handle censored switch times, and simplified likelihood for respondents who do not currently hold a credit card.

second-choice bank A . Treating these as arising from a single draw of latent Gumbel preference shocks — so that the four responses are internally consistent across scenarios — the likelihood contribution is the joint probability of the four stated choices, integrated over the unobserved potential consideration sets:

$$L^{\text{hyp}}(\theta_i) = \sum_{C \in 2^{\mathcal{J}}} P(V, W, R, A \mid C, \theta_i) P(C \mid x^*, \theta_i). \quad (15)$$

We observe whether each respondent would switch under the price shock, the unbundling shock, both, or neither. A price-sensitive consumer (high α_{ik}) may switch when the price worsens but stay when only the complementary product is removed; a consumer who values consolidation (high χ_{ik}^u) may do the opposite. Heterogeneity in $(\alpha_{ik}, \chi_{ik}^u)$ allows the model to rationalize both patterns simultaneously. Appendix D derives the regime-conditional probabilities.

The complete-data log-likelihood factors into the three blocks:

$$\log L(X_i \mid \theta_i) = \log L^{\text{stat}}(\theta_i) + \log L^{\text{dyn}}(\theta_i) + \log L^{\text{hyp}}(\theta_i).$$

The three blocks operate on different timescales but share the consumer-level θ_i , so each block contributes independent identifying variation for the same underlying type.

V.D Algorithm

We maximize the marginal likelihood by Monte Carlo EM, following Train (2008) with one refinement. The E-step forms a posterior over each consumer’s type θ_i by drawing from a proposal distribution and reweighting by importance ratios; the M-step updates the population mean and covariance as weighted first and second moments of the simulated draws.

We depart from Train (2008) in one detail: rather than drawing from the population prior, we draw each consumer’s types from their own posterior from the previous iteration. This observation-specific proposal concentrates draws in the region already judged likely for that consumer, which sharply reduces importance-weight variance and allows the E-step to converge with a manageable number of draws per observation.¹⁷

The Prolific sample identifies consumer-level heterogeneity well but is too small to pin down each bank’s mean utility precisely. After the MC-EM iterations converge, we refine the bank-specific intercepts $\bar{v}_{jk}^u$ to match aggregate market shares from the

¹⁷Appendix D.7 provides full implementation details, including the parametric block structure of Σ_θ , the phased estimation schedule, the adaptive proposal covariance, and the effective-sample-size criterion that triggers re-simulation.

larger MRI-Simmons cross-section. Holding all other parameters fixed at their EM estimates, we iterate $\bar{v}_{jk}^u$ until the model-implied steady-state share for each bank-product cell equals the corresponding MRI share. The refinement adjusts only the bank-specific levels; the consumer-level parameters identified from Prolific are unchanged.

V.E Estimation Results

We estimate the model separately for credit-card transactors and revolvers. Table 4 presents the population means of the structural parameters, Table 6 the bank-specific preference intercepts, and Table 5 the estimated standard deviations and correlations of Σ_θ .¹⁸ We organize the discussion around three parts: price sensitivity, cross-holding mechanisms, and switching frictions.

Price Sensitivity The price sensitivity parameters $\bar{\alpha}_k$ represent the semi-elasticity of choice with respect to a one-percentage-point change in rates. For deposit accounts, both segments exhibit elevated sensitivities ($\bar{\alpha}_1^{\text{transactor}} = 2.25$, $\bar{\alpha}_1^{\text{revolver}} = 1.68$), implying that a one-percentage-point increase in the deposit rate raises the probability of choosing a bank by roughly 1.7–2.3%, conditional on consideration. These estimates lie in the upper range of the empirical literature on deposit demand, which spans semi-elasticities of roughly 0.8–2.3.¹⁹ Within each segment, heterogeneity in α_{ik} is substantial: the log-scale standard deviation of α_{i1} is 1.04 for transactors and 1.40 for revolvers.

The two segments differ sharply on credit cards. Transactors, who pay in full and optimize over rewards, are highly responsive to the reward rate ($\bar{\alpha}_2^{\text{transactor}} = 2.71$). Revolvers, who borrow and face higher switching barriers such as credit-risk-based underwriting at alternative issuers, are much less sensitive to APR ($\bar{\alpha}_2^{\text{revolver}} = 0.49$), a roughly fivefold difference. The estimated variance of α_{i2} is small for both segments.

Cross-holding Mechanisms Both causal channels — consideration spillovers and utility complementarity — are active and quantitatively important. Consideration spillovers are large. For transactors and revolvers alike, an existing credit-card relationship substantially increases the probability of considering the same bank for a deposit account ($\bar{\chi}_1^{c,\text{transactor}} = 1.23$, $\bar{\chi}_1^{c,\text{revolver}} = 0.85$). The deposit relationship has an even stronger impact on credit-card consideration ($\bar{\chi}_2^{c,\text{transactor}} = 1.53$, $\bar{\chi}_2^{c,\text{revolver}} = 1.02$). The boost is large but insufficient on its own to close the visibility gap between a super-regional bank and

¹⁸Standard errors are omitted from the structural tables; parametric bootstrap inference is under development.

¹⁹Reported values include 1.12 in d’Avernas et al. (2023), 1.39 (insured) and 2.26 (uninsured) in Koont (2023), 0.97 in Wang et al. (2022), 0.78 in Dick (2008), and approximately 2.10 in Honka et al. (2017).

a major credit-card issuer. Among transactors, for instance, adding the +1.53-unit boost to the super-regional bank’s baseline credit-card consideration index of -3.42 still falls short of Capital One’s baseline of -1.19 . Even though the mean boost does not fully close the gap, cross-selling expands the super-regional bank’s effective reach.

Consumers also derive substantial utility benefits from consolidating both products at the same bank. For deposit accounts, the utility boost from cross-holding is equivalent to a 35–43 basis-point increase in the deposit rate, computed as $\bar{\chi}_1^u / \bar{\alpha}_1$. For credit cards, the price-equivalent value is 24 basis points of rewards for transactors and a substantially larger 183 basis points of APR for revolvers. The gap likely reflects that revolvers face higher underwriting barriers and lower financial sophistication, which raises the relative value of consolidating both products at their primary deposit bank. We also document substantial heterogeneity in cross-holding utility across consumers in both segments.

Correlated consideration and preferences also contribute to cross-holding. The within-bank consideration correlation $\rho(\nu_1^c, \nu_2^c) = 0.96$ is high across segments: a consumer pre-disposed to consider a bank for checking is much more likely to consider it for credit cards as well. The utility correlation $\rho(\nu_1^u, \nu_2^u)$ is more heterogeneous — 0.65 for transactors but only 0.15 for revolvers — suggesting that transactors’ preferences for a bank’s products are more tightly linked across product lines, while revolvers’ credit card utility is largely independent of their checking preferences.

Switching Frictions Consumers reconsider their products infrequently. The estimated average arrival rates imply reconsideration horizons of about 3.3 to 4.5 years for checking accounts ($\bar{\lambda}_1^{\text{transactor}} = 0.22$, $\bar{\lambda}_1^{\text{revolver}} = 0.30$) and 3.6 to 4.8 years for credit cards ($\bar{\lambda}_2^{\text{transactor}} = 0.21$, $\bar{\lambda}_2^{\text{revolver}} = 0.28$). We also find evidence for persistent preference heterogeneity: the standard deviation of ν_{ijk}^u ranges from 1.33 to 2.01 across products and segments. Long observed tenures thus reflect both infrequent reconsideration and a propensity to remain at the preferred bank upon reconsideration.

V.F Goodness of Fit

To assess the model’s performance, we compare predicted market shares and cross-holding patterns against observed data. Figure 6 shows that our model closely replicates the market shares for both checking accounts and credit cards, capturing the dominance of national players and the long tail of smaller banks. Figure 7 shows that the model matches both the aggregate level of cross-holding and cross-bank variation in cross-holding intensity.

Figure 8 shows that the model reproduces the empirical tenure distributions from the

Prolific survey and the NHCS, correctly matching the heavy right tail of long-tenured consumers.²⁰ Appendix Figure A.5 confirms that this match extends to life-cycle patterns. We simulate synthetic cohorts under the estimated model and find that the model replicates the monotone age gradient in survey tenure for both products, even though λ_{ik} is estimated only from the behavior of existing account holders.

V.G Supply-Side Cost Recovery and Margin Validation

We recover marginal costs at the bank-product level by inverting the supply-side first-order conditions (Equation 12), assuming observed prices reflect a steady-state equilibrium. Under consumer heterogeneity, the first-order conditions remain linear in costs, reducing cost recovery to a 3×3 linear system per bank — one pooled deposit FOC and two segment-specific credit-card FOCs — that we solve in closed form.²¹

The FOCs require several calibrated inputs. We set the discount rate to $r = 0.02$, matching long-run real Treasury yields. We calibrate the balance parameters b_k as the simple average of median values from the SCF, NHCS, and MRI, separately by consumer type. For revolvers, this yields deposits of \$4,620 and credit-card balances of \$3,430; for transactors, deposits of \$22,736 and an annual spending of \$14,247. Appendix C provides further details on balance calibration.

Implied Margins and External Validation Table 7 summarizes the implied margins by product and segment, providing a supply-side validation of the demand estimates against external benchmarks. The median checking account margin of 0.83 percentage points aligns with the deposit spreads documented by Drechsler et al. (2017) and the markups estimated by Wang et al. (2022).

Credit-card margins differ sharply by consumer type. The median transactor margin is a thin 0.45 percentage points, consistent with Federal Reserve profitability reports showing that interchange revenue is largely offset by rewards costs.²² In contrast, the median revolver margin is 4.3 percentage points, driven by interest income on outstanding balances. This is consistent with Nelson (2018) and Drechsler et al. (2025), who find substantial markups in credit-card lending and identify interest income as the primary driver of issuer profitability.

²⁰Appendix D.5 describes how we derive these model-implied tenure distributions by averaging phase-type distributions over the type distribution.

²¹The linearity holds because bank j 's flow payoff vector is affine in its own costs, and the resolvents and their price derivatives depend only on demand parameters and observed prices. Appendix D.7 provides the derivation.

²²Board of Governors of the Federal Reserve System, "Profitability of Credit Card Operations of Depository Institutions," 2023.

VI Counterfactuals

Complementarity reshapes how we assess competition and regulation in retail banking. We begin by quantifying the contribution of complementarity to bank franchise values and its implications for equilibrium pricing. We then conduct three counterfactual exercises — narrow-bank entry, interchange-fee regulation, and a horizontal merger — and compare predictions under the full model to those under a restricted specification that shuts down complementarity ($\chi = 0$). The gap between the two reveals the blind spot of a single-market analyst who treats bank products as independent.

VI.A Loss-Leader Pricing and Bank Franchise Value

Product complementarity contributes substantially to bank franchise values, and banks shade margins below single-product optima to harvest those cross-product gains. We establish the franchise-value stakes first and then show how equilibrium pricing encodes them.

Figure 9 presents two perspectives on how deposit accounts, transactor credit cards, and revolver credit cards contribute to banks' profits. Panel A shows a standard accounting decomposition of each bank's equilibrium profits across the three product lines. For deposit-oriented banks, such as Chase, Bank of America, and Wells Fargo, deposit accounts generate the majority of profits. Capital One and Citi derive a larger share from revolving credit-card balances, reflecting their credit-heavy business models. Transactor credit cards usually contribute a small share, owing to their thin standalone margins.

Panel B measures the marginal contribution of each product line by simulating its removal and recording the resulting decline in total franchise value. For every bank, these contributions sum to well above 100%. The excess arises because removing any line destroys not only its direct profits but also the demand spillovers it generates into complementary products — precisely the complementarity value that Panel A's accounting misses.

Table 8 reports the underlying values. Most starkly, aggregating across the named banks, transactor credit cards account for roughly 16% of accounting profits but contribute about 26% of franchise value — which explains why banks promote them aggressively despite thin standalone margins. The gap is largest at deposit-heavy banks, where removing rewards cards would erase a franchise-value share roughly three times their accounting contribution. Standard accounting radically understates rewards cards' contribution because most of the profit they generate lands on the deposit ledger. The intuition is visible in balance levels: median total deposits are \$23.5k for transactors versus \$3.7k for revolvers (Table 1, Panels C1–C2). Acquiring a cross-holding transac-

tor unlocks a deposit-side margin large enough to cover generous rewards on a card that barely breaks even; revolvers' deposit balances are too thin for the same strategy, and banks instead earn the bulk of the relationship's value from the APR spread on the \$3.4k revolving balance. Rewards cards are thus not standalone profit centers; they are acquisition and retention tools for the higher-margin deposit franchise.

Banks exploit this complementarity through loss-leader pricing. Rearranging the supply-side first-order conditions decomposes each equilibrium margin into two components:

$$\mu_{jk} = \underbrace{\frac{e_j[k]}{A_j[k, k]}}_{\text{own-product markup } \mu_{jk}^{\text{own}}} + \underbrace{-\frac{1}{A_j[k, k]} \sum_{m \neq k} A_j[k, m] \mu_{jm}}_{\text{cross-selling adjustment } \mu_{jk}^{\text{cross}}} . \quad (16)$$

The own-product markup μ_{jk}^{own} is the margin a single-product bank would charge. The cross-selling adjustment μ_{jk}^{cross} captures the off-diagonal terms that a multi-product bank internalizes, each weighted by the complementary product's margin μ_{jm} . A negative cross-selling adjustment means banks shade margins below single-product optima to subsidize customer acquisition on the complementary product.

Figure 10 plots this decomposition for each bank and product. Across essentially all bank-product pairs, the cross-selling adjustment is negative: equilibrium margins are systematically below their single-product counterparts. The gap is largest on transactor credit cards at megabanks, where the own-product markup is roughly one-third larger than the equilibrium margin — confirming that large banks price rewards cards well below what a standalone issuer would charge. This loss-leader strategy allows banks to attract new customers and subsequently earn higher margins by cross-selling them into the deposit franchise.

VI.B Narrow Bank Entry

A long-standing puzzle in retail banking is the relative scarcity of entry in the deposit market despite persistently high spreads (Drechsler et al., 2017). Our results offer one potential explanation: product complementarity creates substantial entry barriers for single-product entrants.

We simulate the entry of a narrow bank that offers a deposit account at a 3% interest rate, equivalent to the federal funds rate and well above the near-zero rates offered by most incumbents. We set the entrant's consideration probability at the median of existing banks and allow incumbents to reoptimize all prices to the new Nash equilibrium. Under our full model, the narrow bank captures 18% of the deposit market, compared

with 23% under the $\chi = 0$ benchmark. Complementarity insulates incumbents: cross-holding consumers stay with their multi-product bank even when the entrant offers a substantially higher deposit rate.

Figure 11, Panel A shows the deposit-share impact on each incumbent. Every multi-product bank retains more deposit share under the full model than under $\chi = 0$, and the protection is largest for banks with strong credit-card franchises, where cross-selling ties bind cardholders more tightly to the deposit side.

Narrow bank entry also spills over into the credit-card market. Panel B shows that incumbents cut credit-card rewards in response: with a smaller deposit base to cross-sell into, the loss-leader strategy would yield lower returns. Credit-dominant banks gain modestly in credit-card share as their deposit-heavy rivals pull back on rewards.

VI.C Regulatory Spillovers

The recent reintroduction of the Credit Card Competition Act (2026) aims to reduce interchange fees by requiring large banks to enable at least two unaffiliated networks for transaction routing. While the primary intent is to lower merchant costs, product complementarity implies that the regulation may propagate well beyond the credit-card market. To assess these cross-market spillovers, we simulate a 1 percentage-point increase in transactor credit-card costs, mimicking the impact of interchange-fee regulation.

Figure 12 shows the impact of the policy. Banks pass through nearly the full cost increase as lower rewards, and the weaker rewards erode the cross-holding ties that funnel cardholders into checking accounts. Banks with large credit-card franchises therefore lose deposit share and respond by raising deposit rates to stem the outflow. Across banks, the correlation between deposit-share losses and changes in credit-card rewards is 0.94, confirming that a fee regulation aimed at the card market propagates through complementarity into the deposit market.

VI.D Horizontal Merger

We simulate a horizontal merger between Citi and the super-regional bank, comparing equilibrium prices under the full model against those under the $\chi = 0$ benchmark. Both brands remain active. The combined firm owns all four products and solves a joint first-order condition, internalizing cross-product complementarity across the combined portfolio. Rivals reoptimize in the new Nash equilibrium.

Under $\chi = 0$, the merger delivers the standard result: reduced competition leads to less consumer-favorable prices, as a conventional concentration analysis would predict (Figure 13, orange markers).

Complementarity inverts the pricing logic. The combined firm exploits cross-selling synergies that neither partner could capture alone, using each brand’s credit card as a loss leader for the other’s deposit base. The brand with the stronger card raises rewards and shades its deposit rate down — harvesting surplus from captive depositors while funneling new cardholders into the partner’s higher-margin deposit franchise. The other brand plays the symmetric role on the deposit side (Figure 13, blue markers). Rival banks respond defensively by raising deposit rates — the opposite of what a standard concentration analysis would predict. The full Nash response runs through credit-card pricing and rivals’ deposit adjustments, not through deposit concentration; a standard deposit-HHI screen misses it entirely.

When merging parties have meaningful cross-product complementarities, joint pricing creates cross-selling gains that two separate firms could not realize — and that deposit concentration metrics cannot detect. Merger analysis in retail banking should thus incorporate cross-holding diagnostics rather than deposit concentration alone.

VII Conclusion

This paper shows that cross-selling and utility complementarity, not correlated preferences alone, drive consumers’ tendency to hold multiple financial products at the same bank. Across three U.S. consumer surveys, roughly 35% of consumers with one checking account and one credit card hold both at the same bank, more than five times the 6.4% benchmark implied by independent product choice; the relationship multiplier averages $6\times$ at megabanks and exceeds $60\times$ at regional banks. Banks exploit consumers’ infrequent reoptimization by using one product as a gateway to capture demand for the other.

A dynamic model of joint deposit and credit card choices, estimated using survey and cross-sectional data, allows us to separately identify these mechanisms and trace their equilibrium implications. These mechanisms reshape standard measures of bank performance. Reward credit cards account for roughly 16% of accounting profits but contribute about 26% of franchise value once their cross-product pull on deposits is accounted for, and rewards-card margins at large banks would be about a third higher absent the cross-product subsidy. Cross-product complementarity creates entry barriers for narrow banks, generates regulatory spillovers from interchange-fee regulations to deposit markets, and creates cross-selling synergies that make horizontal mergers procompetitive.

References

- Agarwal, Sumit, Souphala Chomsisengphet, Chunlin Liu, Changcheng Song, and Nicholas S Souleles**, “Benefits of Relationship Banking: Evidence from Consumer Credit Markets,” *Journal of Monetary Economics*, 2018, 96, 16–32.
- , –, **Neale Mahoney, and Johannes Stroebel**, “Regulating Consumer Financial Products: Evidence from Credit Cards *,” *The Quarterly Journal of Economics*, February 2015, 130 (1), 111–164.
- Aguirregabiria, Victor, Robert Clark, and Hui Wang**, “The geographic flow of bank funding and access to credit: Branch networks, local synergies and competition,” *arXiv preprint arXiv:2407.03517*, 2024.
- Arcidiacono, Peter, Patrick Bayer, Jason R Blevins, and Paul B Ellickson**, “Estimation of Dynamic Discrete Choice Models in Continuous Time with an Application to Retail Competition,” *The Review of Economic Studies*, 2016, 83 (3), 889–931.
- Basten, Christoph and Ragnar Juelsrud**, “Cross-Selling in Bank-Household Relationships: Mechanisms and Implications for Pricing,” *The Review of Financial Studies*, 2023, p. hhad062.
- Berry, Steven, James Levinsohn, and Ariel Pakes**, “Differentiated Products Demand Systems from a Combination of Micro and Macro Data: The New Car Market,” *Journal of Political Economy*, February 2004, 112 (1), 68–105.
- Brown, Zach Y, Mark L Egan, Jihye Jeon, Chuqing Jin, and Alex A Wu**, “Why Do Index Funds Have Market Power? Quantifying Frictions in the Index Fund Market,” Technical Report, National Bureau of Economic Research 2023.
- Caldwell, Sydnee, Ingrid Haegeler, and Jörg Heining**, “Why Workers Stay: Pay, Beliefs, and Attachment,” 2025.
- Cuesta, José Ignacio and Alberto Sepúlveda**, “Price Regulation in Credit Markets: A Trade-off between Consumer Protection and Credit Access,” *Available at SSRN 3282910*, 2021.
- d’Avernas, Adrien, Andrea L Eisfeldt, Can Huang, Richard Stanton, and Nancy Wallace**, “The deposit business at large vs. small banks,” Technical Report, National Bureau of Economic Research 2023.
- Dick, Astrid A**, “Demand estimation and consumer welfare in the banking industry,” *Journal of Banking & Finance*, 2008, 32 (8), 1661–1676.
- Drechsler, Itamar, Alexi Savov, and Philipp Schnabl**, “The deposits channel of monetary policy,” *The Quarterly Journal of Economics*, 2017, 132 (4), 1819–1876.
- , –, **and** –, “Banking on Deposits: Maturity Transformation without Interest Rate Risk,” *The Journal of Finance*, 2021, 76 (3), 1091–1143.

- , Hyeyoon Jung, Weiyu Peng, Dominik Supera, and Guanyu Zhou, “Credit Card Banking,” *Working Paper*, March 2025.
- Economides, Nicholas and Steven C Salop**, “Competition and integration among complements, and network market structure,” *The Journal of Industrial Economics*, 1992, pp. 105–123.
- Egan, Mark, Ali Hortaçsu, and Gregor Matvos**, “Deposit Competition and Financial Fragility: Evidence from the Us Banking Sector,” *American Economic Review*, 2017, 107 (1), 169–216.
- Egan, Mark L., Ali Hortaçsu, Nathan A. Kaplan, Adi Sunderam, and Vincent Yao**, “Dynamic Competition for Sleepy Deposits,” September 2025.
- Galenianos, Manolis and Alessandro Gavazza**, “Regulatory Interventions in Consumer Financial Markets: The Case of Credit Cards,” *Journal of the European Economic Association*, 2022, 20 (5), 1897–1932.
- Gentzkow, Matthew**, “Valuing New Goods in a Model with Complementarity: Online Newspapers,” *American Economic Review*, 2007, 97 (3), 713–744.
- Goeree, Michelle Sovinsky**, “Limited Information and Advertising in the U.S. Personal Computer Industry,” *Econometrica*, 2008, 76 (5), 1017–1074.
- Grieco, Paul LE, Charles Murry, and Ali Yurukoglu**, “The evolution of market power in the us automobile industry,” *The Quarterly Journal of Economics*, 2024, 139 (2), 1201–1253.
- Hastings, Justine, Ali Hortaçsu, and Chad Syverson**, “Sales Force and Competition in Financial Product Markets: The Case of Mexico’s Social Security Privatization,” *Econometrica : journal of the Econometric Society*, 2017, 85 (6), 1723–1761.
- Hellmann, Thomas, Laura Lindsey, and Manju Puri**, “Building Relationships Early: Banks in Venture Capital,” *The Review of Financial Studies*, 2008, 21 (2), 513–541.
- Honka, Elisabeth, Ali Hortaçsu, and Maria Ana Vitorino**, “Advertising, Consumer Awareness, and Choice: Evidence from the US Banking Industry,” *The RAND Journal of Economics*, 2017, 48 (3), 611–646.
- Hortaçsu, Ali and Chad Syverson**, “Product Differentiation, Search Costs, and Competition in the Mutual Fund Industry: A Case Study of S&P 500 Index Funds,” *The Quarterly journal of economics*, 2004, 119 (2), 403–456.
- Ivashina, Victoria and Anna Kovner**, “The Private Equity Advantage: Leveraged Buy-out Firms and Relationship Banking,” *The Review of Financial Studies*, 2011, 24 (7), 2462–2498.
- JPM**, “JPM 2015 Investor Day.Pdf,” February 2015.

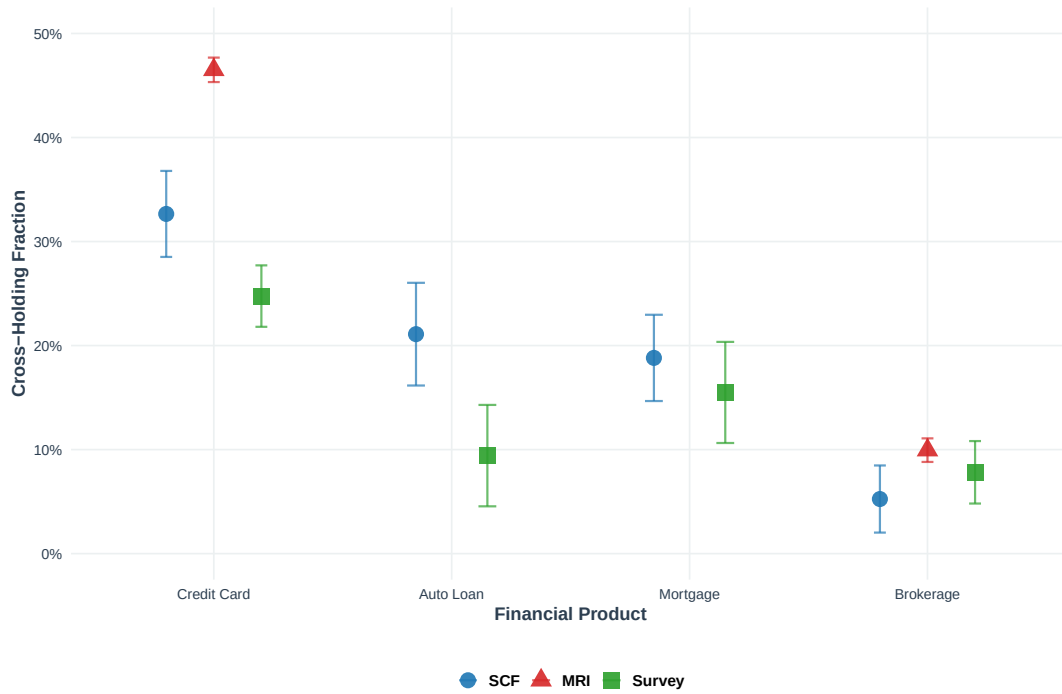
- Koont, Naz**, "The digital banking revolution: Effects on competition and stability," *Available at SSRN 4624751*, 2023.
- Laux, Christian and Uwe Walz**, "Cross-Selling Lending and Underwriting: Scope Economies and Incentives," *Review of Finance*, 2009, 13 (2), 341–367.
- Mas, Alexandre and Amanda Pallais**, "Valuing alternative work arrangements," *American Economic Review*, 2017, 107 (12), 3722–3759.
- **and —**, "Labor supply and the value of non-work time: Experimental estimates from the field," *American Economic Review: Insights*, 2019, 1 (1), 111–126.
- Nalebuff, Barry**, "Bundling as an Entry Barrier," *The Quarterly Journal of Economics*, 2004, 119 (1), 159–187.
- Nelson, Scott**, "Private Information and Price Regulation in the Us Credit Card Market," *Unpublished working paper*, 2018.
- Parlour, Christine A, Uday Rajan, and Haoxiang Zhu**, "When FinTech Competes for Payment Flows," *The Review of Financial Studies*, November 2022, 35 (11), 4985–5024.
- Petrin, Amil**, "Quantifying the Benefits of New Products: The Case of the Minivan," *Journal of Political Economy*, 2002, 110 (4).
- Puri, Manju, Jörg Rocholl, and Sascha Steffen**, "What Do a Million Observations Have to Say about Loan Defaults? Opening the Black Box of Relationships," *Journal of Financial Intermediation*, July 2017, 31, 1–15.
- Qi, Yingjie**, "Big Broad Banks: How Does Cross-Selling Affect Lending?," *Review of Finance*, 2024, 28 (2), 551–592.
- Santikian, Lori**, "The Ties That Bind: Bank Relationships and Small Business Lending," *Journal of financial intermediation*, 2014, 23 (2), 177–213.
- Song, Minjae, Sean Nicholson, and Claudio Lucarelli**, "Mergers with interfirm bundling: a case of pharmaceutical cocktails," *The RAND journal of Economics*, 2017, 48 (3), 810–834.
- Stango, Victor and Jonathan Zinman**, "Borrowing High versus Borrowing Higher: Price Dispersion and Shopping Behavior in the US Credit Card Market," *The Review of Financial Studies*, 2016, 29 (4), 979–1006.
- Train, Kenneth E.**, "EM Algorithms for Nonparametric Estimation of Mixing Distributions," *Journal of Choice Modelling*, 2008, 1 (1), 40–69.
- U.S. Department of Justice**, "Bank Merger Competitive Review – Introduction and Overview," 1995.
- U.S. Department of Justice and Federal Trade Commission**, "Horizontal Merger Guidelines," 2023.

Wang, Lulu, “Regulating Competing Payment Networks,” URL: https://luluywang.github.io/PaperRepository/payment_jmp.pdf (last accessed on 10 November 2023), 2023.

Wang, Yifei, Toni M. Whited, Yufeng Wu, and Kairong Xiao, “Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation,” *The Journal of Finance*, 2022, 77 (4), 2093–2141.

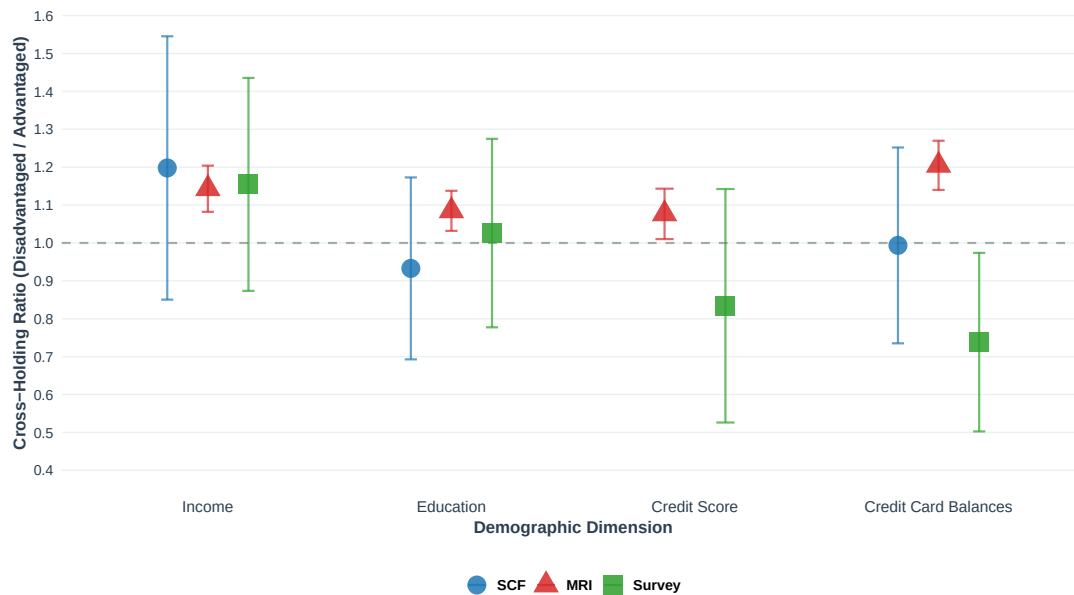
Whinston, Michael D, “Tying, Foreclosure, and Exclusion,” 1989.

Figure 1: Cross-holding of Checking Accounts and Other Products



Notes: The fraction shown reflects the share of customers holding additional products at their primary bank. For each product, we calculate the fraction among consumers holding both a checking account and that product.

Figure 2: Comparing Cross-holding of Disadvantaged versus Advantaged Groups



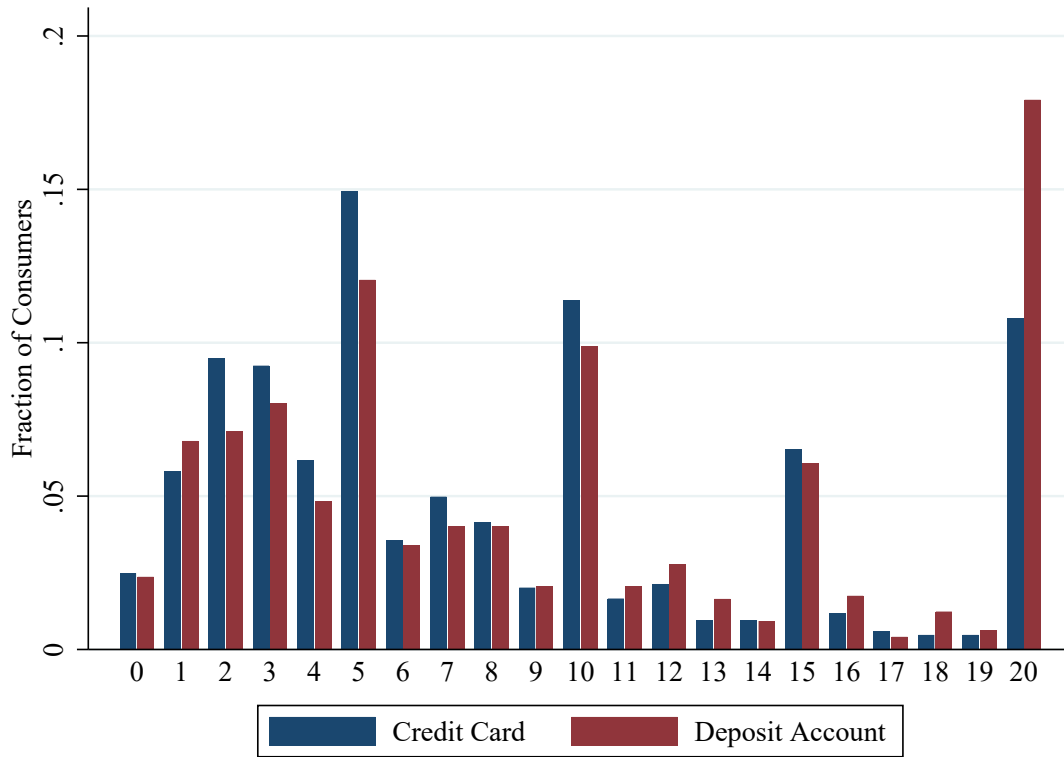
Notes: The figure compares cross-holding between disadvantaged and advantaged consumer groups. We define the disadvantaged group as those with low income, low education, low credit scores, or high credit card balances (i.e., carrying a balance sometimes or often). Specifically, the lower education group consists of consumers with a high school degree or below, and low credit scores are those below 720. Income is segmented using the sample median. We proxy revolvers using the balance carrier group (those who carry a balance sometimes or often) in contrast to transactors, who spend to collect rewards.

Figure 3: Cross-holding of Checking Accounts and Credit Cards by Bank



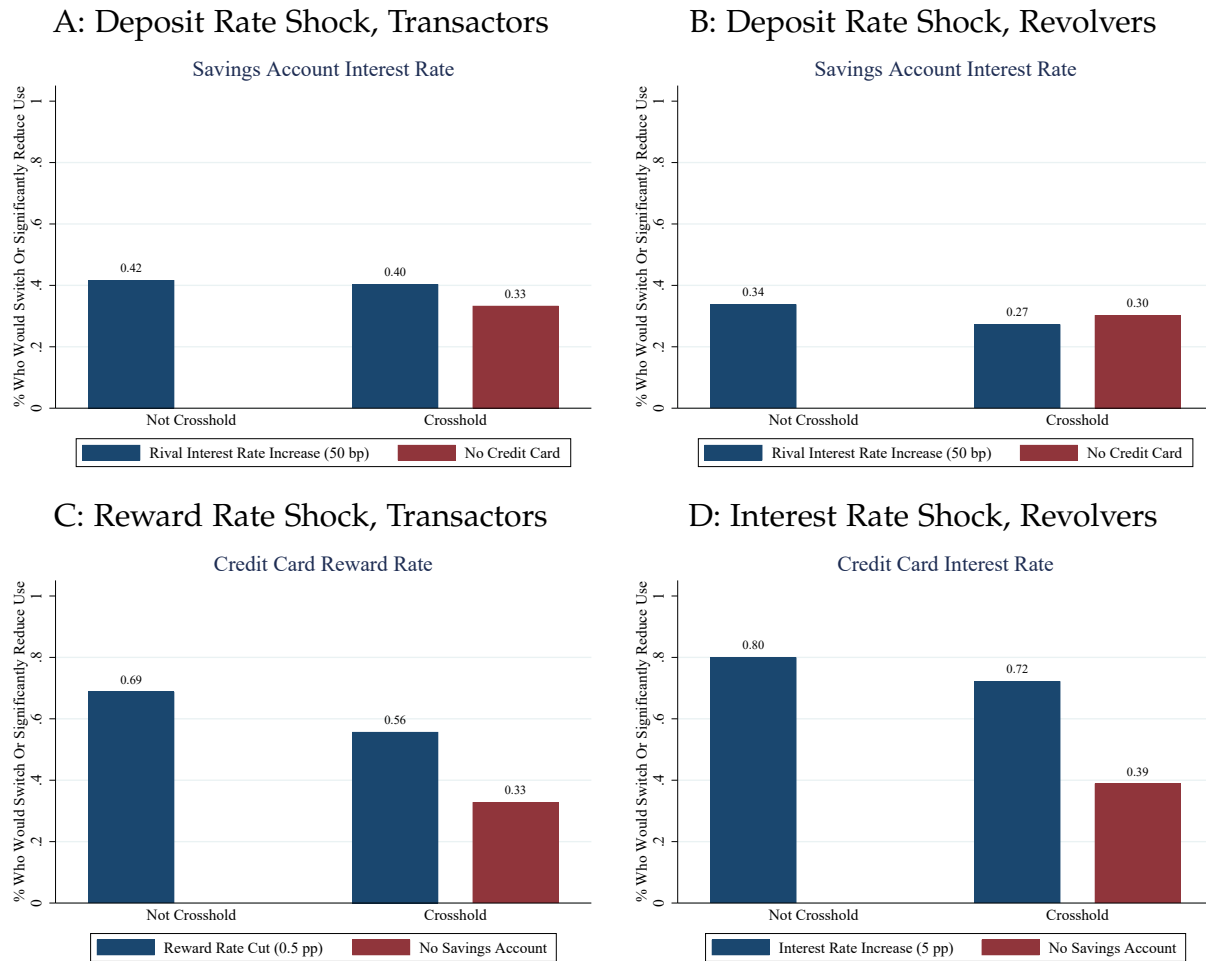
Notes: This figure shows the prevalence of cross-holding between checking accounts and credit cards. The top panel shows each bank's checking account market share among two groups of consumers: those who hold a credit card at the bank (blue dots) and those who do not (orange dots). The bottom panel shows the credit card market shares among consumers who do and do not have a checking account at the bank. Market shares reflect both primary and other accounts. The sample includes respondents in the 2021 and 2022 MRI data.

Figure 4: Consumer Tenure at Current Banks



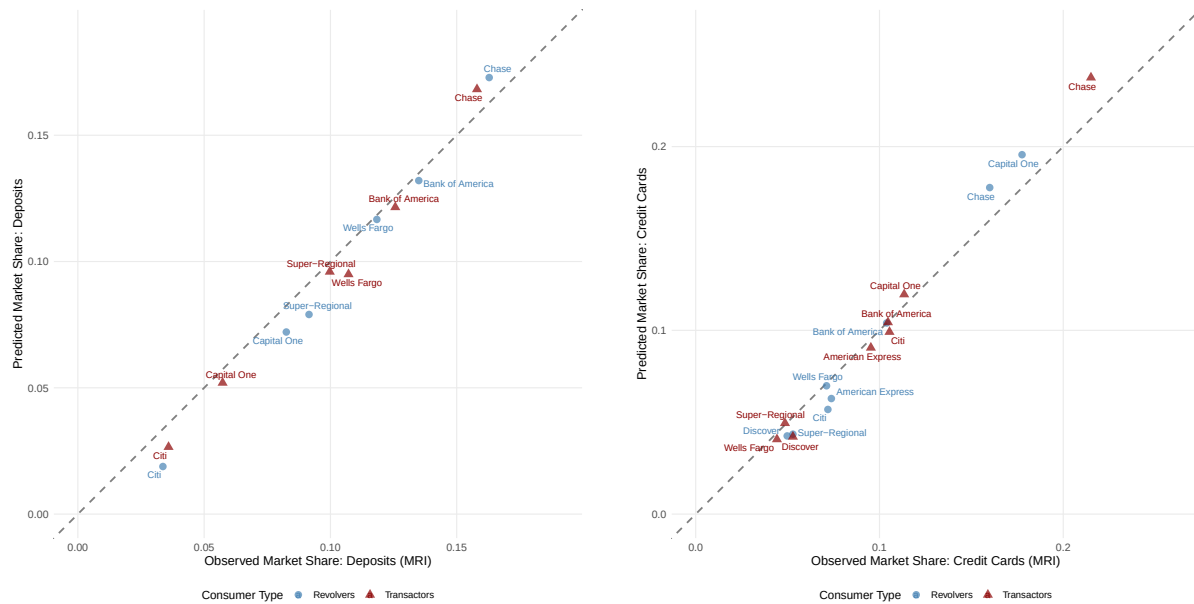
Notes: This figure shows the distribution of consumer tenure at their current bank for credit account and deposit account, separately. 0 means the consumer obtained their current product less than a year ago, while 20 means more than twenty years ago.

Figure 5: Stated Responses to Hypothetical Shocks



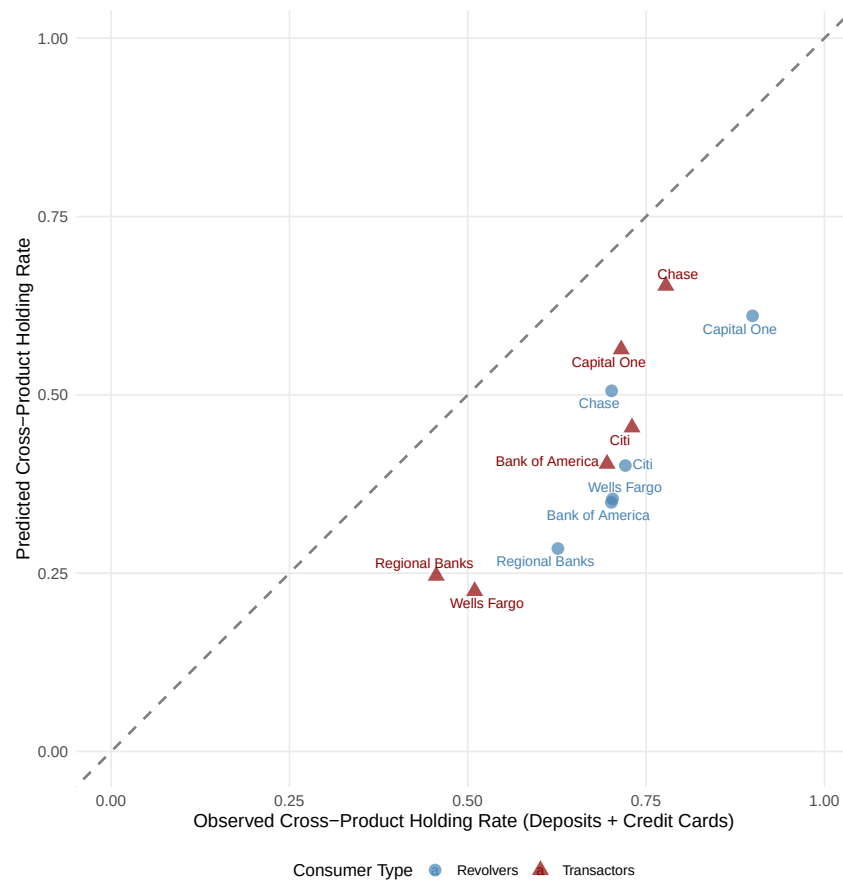
Notes: This figure shows the fraction of consumers who state they would switch to a different bank or significantly reduce the usage of their current product under various hypothetical shocks. Transactors are individuals who never or rarely carry a balance on their credit cards, while revolvers are individuals who carry a balance at least occasionally. The “crosshold” group consists of consumers who currently hold their primary deposit account and credit card at the same bank.

Figure 6: Goodness of Fit: Market Shares



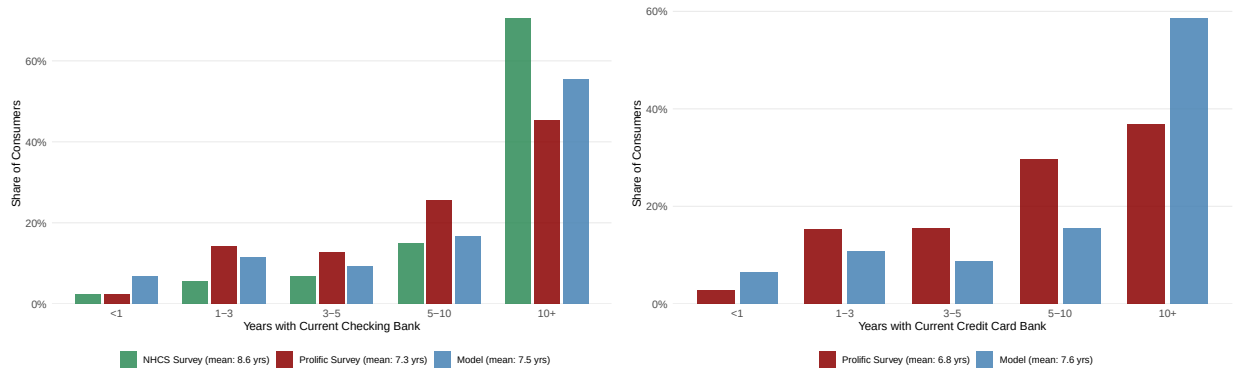
Notes: These figures compare observed and model-predicted market shares for checking accounts (left) and credit cards (right). Each point represents a bank–segment pair (transactor and revolver shown separately, distinguished by color and shape), with the horizontal axis showing observed shares and the vertical axis showing predicted shares. Points along the 45-degree line indicate perfect fit.

Figure 7: Goodness of Fit: Cross-holding Rates by Bank and Consumer Type



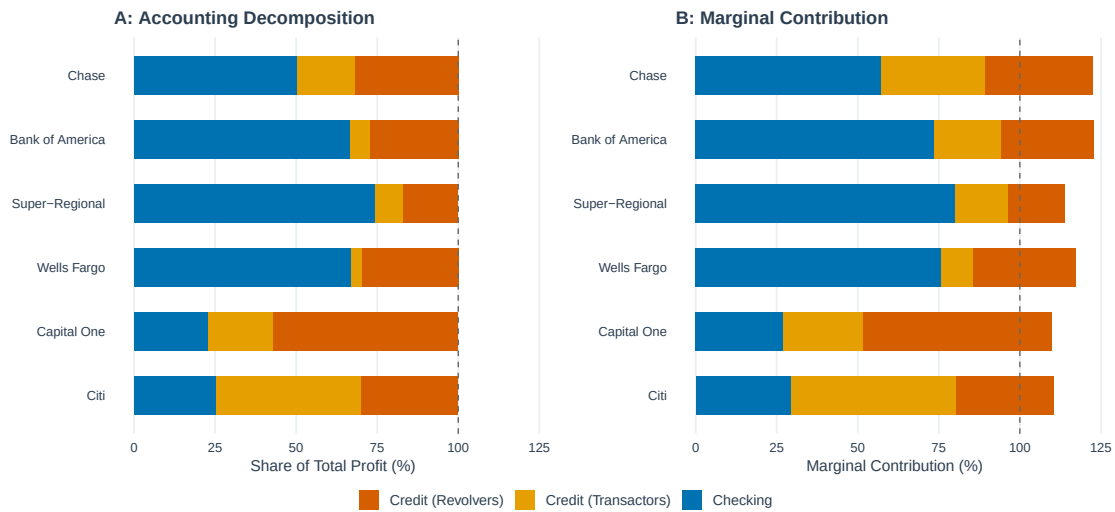
Notes: This figure compares observed and model-predicted cross-holding rates across banks for transactors and revolvers. The figure shows the probability of holding a credit card at a bank conditional on having a checking account there.

Figure 8: Goodness of Fit: Tenure Distributions



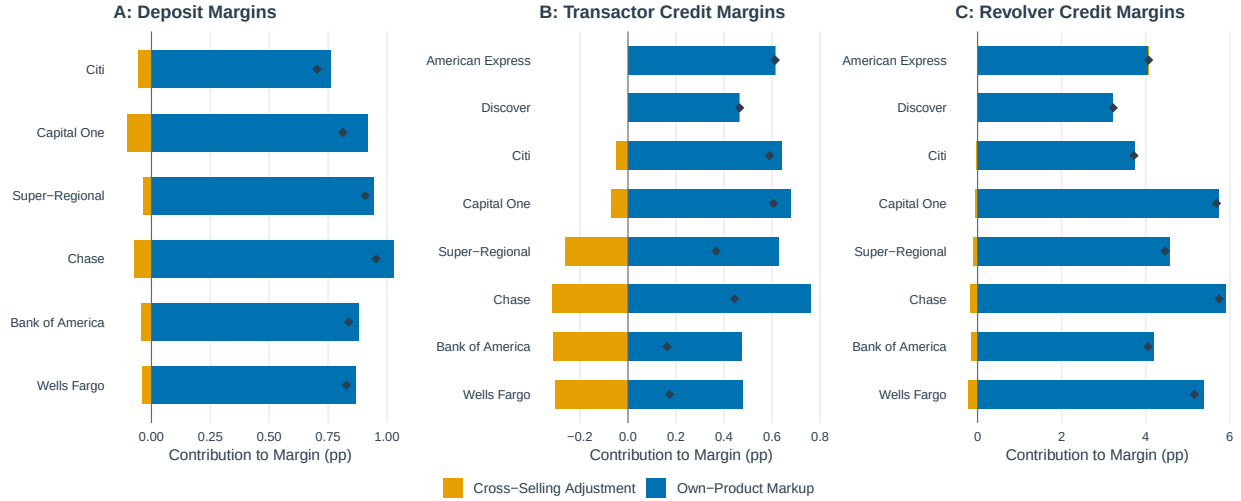
Notes: These figures compare model-predicted and observed tenure distributions for checking accounts (left) and credit cards (right). Tenures are binned into five categories: less than 1 year, 1–3 years, 3–5 years, 5–10 years, and 10+ years. The model predictions are computed from the backward recurrence time distribution of the estimated CTMC (see Appendix D.5). Observed tenures come from the Prolific survey, with NHCS survey data shown as an additional benchmark for checking accounts.

Figure 9: Profit Decomposition and Franchise Values



Notes: This figure decomposes bank franchise values across product lines. Panel A shows the accounting decomposition: each bank's total profit is split into contributions from checking accounts, credit card transactors, and credit card revolvers. Panel B shows the marginal contribution of each product line, computed by removing that product and measuring the resulting loss in the bank's total franchise value. The dashed vertical line marks 100%. Marginal contributions sum to more than 100%, reflecting cross-product complementarities: removing any single product line reduces profits from the remaining lines as well.

Figure 10: FOC Margin Decomposition: Own-Product Markup vs. Cross-Selling Adjustment



Notes: This figure decomposes each bank's equilibrium margin (percentage points) into two components. The own-product markup ($\mu_{jk}^{\text{own}} = e_j[k] / A_j[k, k]$) is the margin a single-product optimizer would set; the cross-selling adjustment ($\mu_{jk}^{\text{cross}} = -A_j[k, k]^{-1} \sum_{m \neq k} A_j[k, m] \mu_{jm}$) is the margin shading a multi-product bank applies to internalize complementary-product profits (see Equation 16). Panels A–C show deposits, transactor credit, and revolver credit, respectively. Diamonds mark the total margin μ_{jk} . A negative cross-selling adjustment indicates that the bank prices product k more generously than a standalone competitor would, in order to harvest profits on the complementary product line.

Figure 11: Counterfactual: Narrow Bank Entry



Notes: This figure shows equilibrium outcomes under the entry of a narrow bank that offers only a deposit account. Panel A shows market shares, Panel B shows prices, and Panel C shows profits under varying levels of cross-product complementarity.

Figure 12: Counterfactual: Interchange Fee Regulation



Notes: This figure shows equilibrium outcomes under interchange fee regulation that reduces credit card reward rates. Panel A shows market shares, Panel B shows prices, and Panel C shows profits. The results illustrate regulatory spillovers from the credit card market to the deposit market.

Figure 13: Counterfactual: Horizontal Merger (Citi + Super-Regional Bank)



Notes: This figure shows equilibrium outcomes under a horizontal merger between Citi and the super-regional bank. Both brands remain active in the choice set; what changes is that the combined firm now owns all four products (checking and credit at each brand) and solves a joint first-order condition over them, while cross-selling complementarity χ pools across the firm so the bonus fires for any pair of products held within the combined entity. Panel A shows post-merger changes in credit card and deposit market shares by bank; Panel B shows changes in credit card reward rates and deposit rates. Results are reported for the full model (blue) and the restricted model that shuts down cross-product complementarity, $\chi = 0$ (orange). Under the restricted model, joint pricing alone yields no synergy: the combined firm loses both deposit and credit share to rivals and profits decline. Under the full model, the broader span of complementarity within the firm makes the merger profitable: Citi lowers its deposit rate to extract surplus while raising credit-card rewards to acquire customers, and combined profits rise by 4.3%.

Table 1: Summary Statistics

Characteristic	Data Source			
	Prolific	MRI	NHCS	SCF
<i>Panel A: Demographics</i>				
Median Income (in thousand dollars)	62.0	67.1	67.1	70.5
Bachelor's Degree or Above (%)	53.9	35.9	34.7	40.1
Credit Score ≥ 720 (%)	77.8	78.1		
Carrying Credit Card Balance (%)	43.3	36.6	43.4	39.3
<i>Panel B: Product Holdings (%)</i>				
Deposit Account (%)	98.8	94.4	94.1	94.6
Credit Card (%)	86.0	76.0	73.1	80.0
Auto Loan (%)	22.5	24.3	31.4	34.1
Mortgage (%)	30.3	41.3	32.0	40.9
Brokerage Account (%)	32.6	32.5	18.9	26.0
<i>Panel C1: Account Balances - Revolvers (in thousands)</i>				
Median Total Deposits (k\$)	1.1		5.5	3.7
Median Credit Card Balance (k\$)	0.9			3.4
Median Credit Card Spending (k\$/yr)	2.8	4.1		3.8
<i>Panel C2: Account Balances - Transactors (in thousands)</i>				
Median Total Deposits (k\$)	9.0		22.0	23.5
Median Credit Card Spending (k\$/yr)	8.4	10.2		18.4
Observations	955	100,884	39,716	4,595

Notes: This table reports summary statistics across our main data sources. Prolific refers to our original survey of approximately 1,000 U.S. consumers. MRI refers to the 2021–2022 waves of the MRI-Simmons USA Study. NHCS refers to the 2020 wave of the MRI-Simmons USA Study, which additionally captures consumer tenure at their financial institutions. SCF refers to the 2022 Survey of Consumer Finances. Panel A compares demographic characteristics. Panel B compares product ownership rates. Panels C1 and C2 report median account balances for revolvers (consumers who carry credit card balances) and transactors (consumers who pay their balance in full each month), respectively. Cells are blank where the data source does not report the variable. Prolific respondents hold lower median balances than SCF and NHCS because the Prolific panel skews younger and lower-income; the structural price sensitivities estimated on this sample nonetheless match external benchmarks (see Section V), suggesting that the demand parameters identified from this sample generalize.

Table 2: Bank Market Shares in Deposit Accounts and Credit Cards

Bank	Deposit (MRI)	Deposit (Prolific)	Deposit (SOD)	Credit (MRI)	Credit (Prolific)	Credit (Nilson)
Chase	13.6	10.7	11.7	17.4	22.4	17.0
Bank of America	11.1	7.7	11.0	10.6	9.5	8.3
Wells Fargo	10.6	7.7	8.3	6.0	4.5	3.0
Capital One	5.2	10.3	2.2	14.2	24.0	14.1
PNC	3.0	3.4	2.3	1.4	0.5	0.5
Citi	2.9	1.2	4.1	8.6	7.0	12.1
US Bank	2.8	2.9	2.5	2.2	2.1	3.3
TD Bank	1.8	1.4	2.2	0.8	0.8	1.2
Citizens Bank	1.3	1.3	1.5	0.6	0.1	0.2
American Express	-	3.5	0.5	7.5	8.1	9.0
Discover	-	3.3	0.4	4.9	11.6	7.6
Others	47.6	46.4	53.2	25.7	9.4	23.7
Total	100.0	100.0	100.0	100.0	100.0	100.0

Notes: This table shows market shares for deposit accounts and credit cards in the MRI sample and our Prolific Survey sample. Market shares in MRI are weighted by the provided population weights. The "Others" category includes credit unions, internet banks, mutual funds, and various other national or regional banks. For MRI deposit accounts, the "Others" category also includes American Express and Discover. When a consumer holds multiple accounts for a single product, each account is treated as a separate observation in the market share calculations. For comparison, Column 3 shows deposit account market shares based on the Summary of Deposit Data, and Column 6 shows credit card market shares from the Nilson Report.

Table 3: Factors for Banking Product Considerations

Outcome	(1) Deposit	(2) Deposit	(3) Credit	(4) Credit
Credit Used	0.060*** (0.027)	0.063*** (0.025)		
Credit Considered	0.242*** (0.015)	0.235*** (0.014)		
Deposit Used			0.061** (0.026)	0.108*** (0.024)
Deposit Considered			0.257*** (0.017)	0.257*** (0.017)
Constant	0.117*** (0.003)	0.117*** (0.003)	0.137*** (0.004)	0.135*** (0.004)
Observations	11,796	11,796	11,796	11,796
<i>Controlling for:</i>				
Bank FE		✓		✓

Notes: This table shows factors that predict whether a consumer considers a bank when choosing a deposit account (Columns 1 and 2) or a credit card (Columns 3 and 4). *Credit Used* indicates whether the consumer was using the bank's credit card when choosing their deposit account. *Deposit Used* indicates whether the consumer was using the bank's deposit account when choosing a credit card. *** implies significance at 0.01 level, ** 0.05, * 0.1.

Table 4: Structural Parameter Estimates: Transactors vs. Revolvers

Parameter	Transactors	Revolvers
Price Sensitivity		
Checking (α_1)	2.25	1.68
Credit (α_2)	2.71	0.49
Cross-Selling Effects		
Consideration – Checking (χ_1^c)	1.23	0.85
Consideration – Credit (χ_2^c)	1.53	1.02
Utility – Checking (χ_1^u)	0.78	0.73
Utility – Credit (χ_2^u)	0.64	0.90
Implied Benefits		
Checking (pp deposit rate)	0.35	0.43
Credit (pp cashback/APR)	0.24	1.83
Arrival Rates		
Checking (λ_1)	0.22	0.30
Credit (λ_2)	0.21	0.28

Notes: Reports population means θ of the structural parameters from the MC-EM estimator. N reports the Prolific sample sizes; the full likelihood also incorporates approximately 100,000 MRI-Simmons observations via a weighted summand on the steady-state likelihood (see Section V). Standard errors are omitted; parametric bootstrap inference is under development.

Table 5: Estimated Heterogeneity: Standard Deviations and Correlations of Σ_θ

	Transactors	Revolvers
Bank Preference Std. Deviations (Σ_v)		
v_1^c (Check. Consid.)	1.65	1.59
v_2^c (Credit Consid.)	1.56	1.70
v_1^u (Check. Utility)	1.33	1.65
v_2^u (Credit Utility)	1.51	2.01
Bank Preference Correlations (Σ_v)		
$\rho(v_1^c, v_2^c)$	0.96	0.96
$\rho(v_1^u, v_2^u)$	0.65	0.15
Structural Parameter Std. Deviations (Σ_θ, log scale)		
$\log \alpha_1$ (Checking)	1.04	1.40
$\log \alpha_2$ (Credit)	0.11	0.08
$\log \chi_1^u$	1.00	0.57
$\log \chi_2^u$	0.09	0.37
$\log \lambda_1$ (Checking)	0.11	0.16
$\log \lambda_2$ (Credit)	0.10	0.14

Notes: Reports standard deviations and within-bank correlations of the estimated heterogeneity distribution. The Σ_v block captures bank-preference heterogeneity in levels, with strong within-bank correlation across consideration and utility for the same product (near-unit $\rho(v_1^c, v_2^c)$); cross-correlations between consideration (v^c) and utility (v^u) blocks are restricted to zero and omitted from the table. The $(\log \alpha_k, \log \chi_k^u)$ blocks are estimated separately by product (the cross-product joint distribution is not identified by the survey design) and exhibit substantive dispersion; $(\log \lambda_1, \log \lambda_2)$ is jointly normal across products with limited within-segment dispersion. The consideration complementarity χ^c is held homogeneous within consumer segment ($\bar{\chi}_k^c$ identified by Fact 3; variance not identified).

Table 6: Bank Preference Parameters: Transactors and Revolvers

Bank	v_1^c (Check. Consid.)	v_2^c (Credit Consid.)	v_1^u (Check. Utility)	v_2^u (Credit Utility)
Panel A: Transactors				
American Express	—	-1.69	—	-5.84
Bank of America	-2.07	-1.84	-2.88	-4.65
Capital One	-2.46	-1.19	-8.85	-3.83
Chase	-2.05	-0.61	-3.27	-4.42
Citi	-3.71	-1.84	-9.29	-5.81
Regional Banks	-2.71	-3.42	-4.18	-2.30
Discover	—	-1.44	—	-4.90
US Bank	-2.51	-2.77	-2.07	-3.46
Panel B: Revolvers				
American Express	—	-1.85	—	10.20
Bank of America	-2.10	-2.17	-1.68	14.63
Capital One	-2.59	-0.31	-6.12	13.26
Chase	-1.91	-0.81	-1.91	14.50
Citi	-3.69	-2.18	-7.76	10.30
Regional Banks	-2.25	-3.29	-3.56	10.76
Discover	—	-1.12	—	11.42
US Bank	-2.64	-3.34	-1.14	11.02

Table 7: Implied Margins and Costs from Supply-Side First-Order Conditions

Statistic	Value
Margins (Median)	
Median Checking Margin (pp)	0.83
Median Transactor Credit Margin (pp)	0.45
Median Revolver Credit Margin (pp)	4.27
Checking Cost Range	
Min Checking Cost	-0.0369
Max Checking Cost	-0.0202
Transactor Credit Cost Range	
Min Transactor Credit Cost	-0.0278
Max Transactor Credit Cost	-0.0135
Revolver Credit Cost Range	
Min Revolver Credit Cost	0.1549
Max Revolver Credit Cost	0.2226

Notes: This table reports implied margins and marginal costs recovered from the supply-side first-order conditions of the EM specification. Margins are the difference between per-unit revenue and marginal cost for each product, in percentage points, averaged across consumer types. Costs represent the marginal cost of serving a customer per dollar of balance, net of interchange and investment-return revenue; negative values indicate that these net revenues exceed direct servicing costs.

Table 8: Profit Decomposition: Full Model

Bank	Accounting (% of total profit)				Marginal Contribution (% of total profit)			
	Checking	Trans.	Rev.	Total	Checking	Trans.	Rev.	Total
Chase	50.3	17.9	31.8	100.0	57.1	32.2	33.1	122.5
Bank of America	66.7	6.0	27.3	100.0	73.8	20.4	28.7	122.8
Super-Regional	74.4	8.7	16.9	100.0	80.0	16.4	17.5	114.0
Wells Fargo	67.1	3.2	29.7	100.0	75.8	9.7	31.6	117.1
Capital One	22.8	20.2	56.9	100.0	27.0	24.7	58.1	109.9
Citi	25.3	44.9	29.8	100.0	29.3	51.2	30.1	110.5
Average	50.5	15.5	34.0	100.0	56.7	25.6	35.2	117.5

Profit decomposition for the estimated model (χ free). Sample restricted to deposit-and-credit banks; AmEx and Discover are credit-only and excluded. Banks are ordered by predicted checking share (largest first). Accounting Decomposition: each line of business's baseline profit as a share of bank-total baseline profit; rows sum to 100% by construction. Lines of business are checking, credit-card transactors, and credit-card revolvers. Marginal Contribution: change in bank-total profit from removing each line of business holding the others fixed, expressed as % of bank-total baseline profit. The Total column is the sum across the three MCs. MC Total > 100% measures complementarity: removing a product also reduces acquisition and retention of the others, so the joint profit lost exceeds the product's stand-alone accounting share. Under $\chi = 0$ the cross-selling channel is shut down and MC Totals collapse toward 100%. The Average row is the cross-bank mean of each column, weighted by each bank's baseline total profit.

Table 9: Profit Decomposition: Restricted Model ($\chi = 0$)

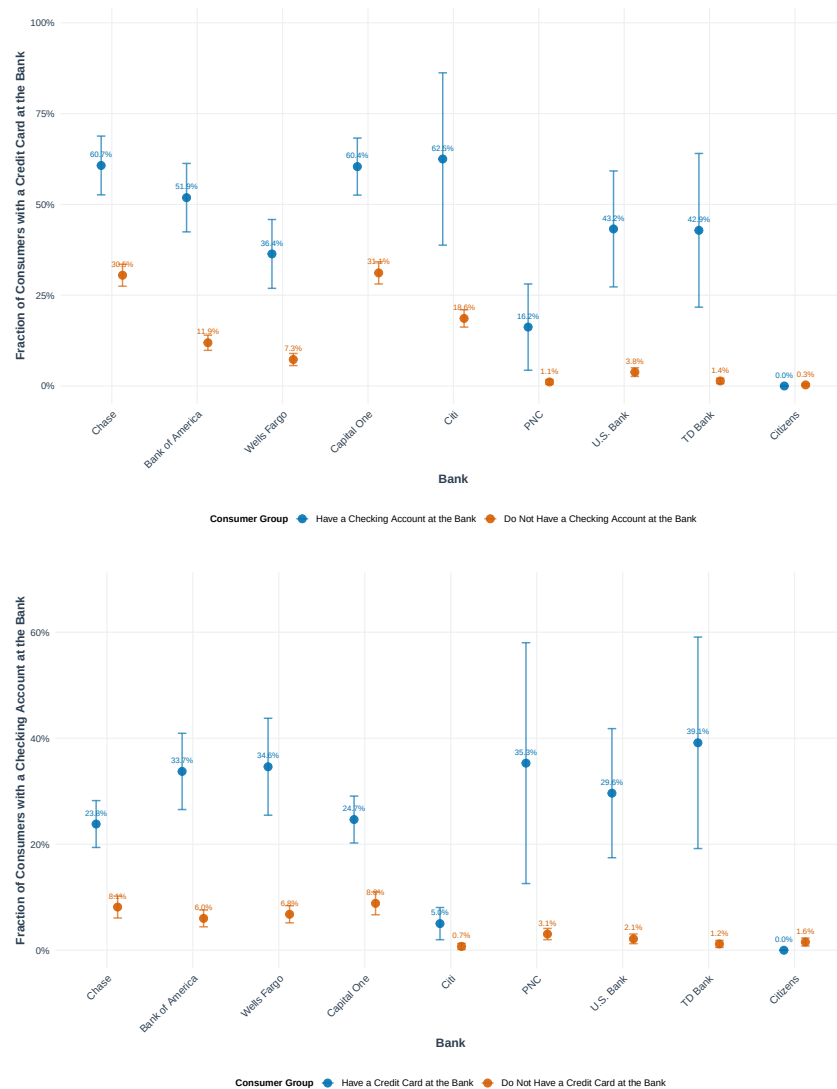
Bank	Accounting (% of total profit)				Marginal Contribution (% of total profit)			
	Checking	Trans.	Rev.	Total	Checking	Trans.	Rev.	Total
Chase	37.7	33.5	28.8	100.0	37.7	33.5	28.8	100.0
Bank of America	51.9	20.6	27.5	100.0	51.9	20.6	27.5	100.0
Super-Regional	67.3	16.2	16.5	100.0	67.3	16.2	16.5	100.0
Wells Fargo	63.8	10.5	25.8	100.0	63.8	10.5	25.8	100.0
Capital One	14.9	24.2	60.9	100.0	14.9	24.2	60.9	100.0
Citi	18.0	51.0	31.0	100.0	18.0	51.0	31.0	100.0
Average	40.5	26.7	32.8	100.0	40.5	26.7	32.8	100.0

Profit decomposition for the restricted model ($\chi = 0$). Sample restricted to deposit-and-credit banks; AmEx and Discover are credit-only and excluded. Banks are ordered by predicted checking share (largest first). Accounting Decomposition: each line of business's baseline profit as a share of bank-total baseline profit; rows sum to 100% by construction. Lines of business are checking, credit-card transactors, and credit-card revolvers. Marginal Contribution: change in bank-total profit from removing each line of business holding the others fixed, expressed as % of bank-total baseline profit. The Total column is the sum across the three MCs. MC Total > 100% measures complementarity: removing a product also reduces acquisition and retention of the others, so the joint profit lost exceeds the product's stand-alone accounting share. Under $\chi = 0$ the cross-selling channel is shut down and MC Totals collapse toward 100%. The Average row is the cross-bank mean of each column, weighted by each bank's baseline total profit.

Appendices. For Online Publication Only

A Additional Figures and Tables

Figure A.1: Crossholding of Checking Accounts and Credit Cards by Bank, Prolific Data



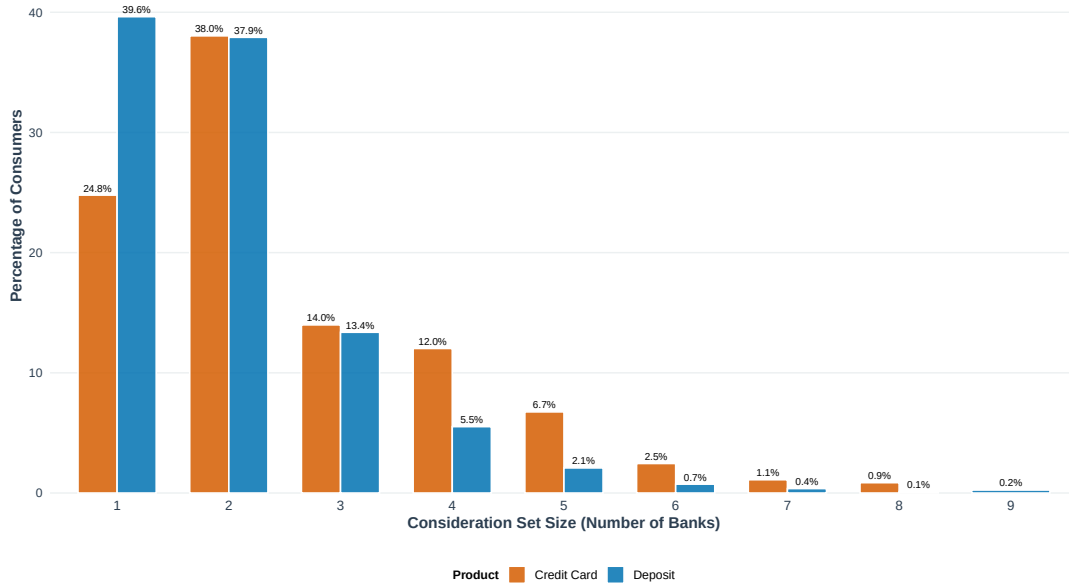
Notes: This figure shows the deposit account market shares of Chase, Wells Fargo, and Bank of America among two groups of consumers: those who have a brokerage service account at the bank (orange dots) and those who do not (green dots). The sample includes all consumers in the 2021 and 2022 MRI data.

Figure A.2: Cross-holding at Big and Small Banks



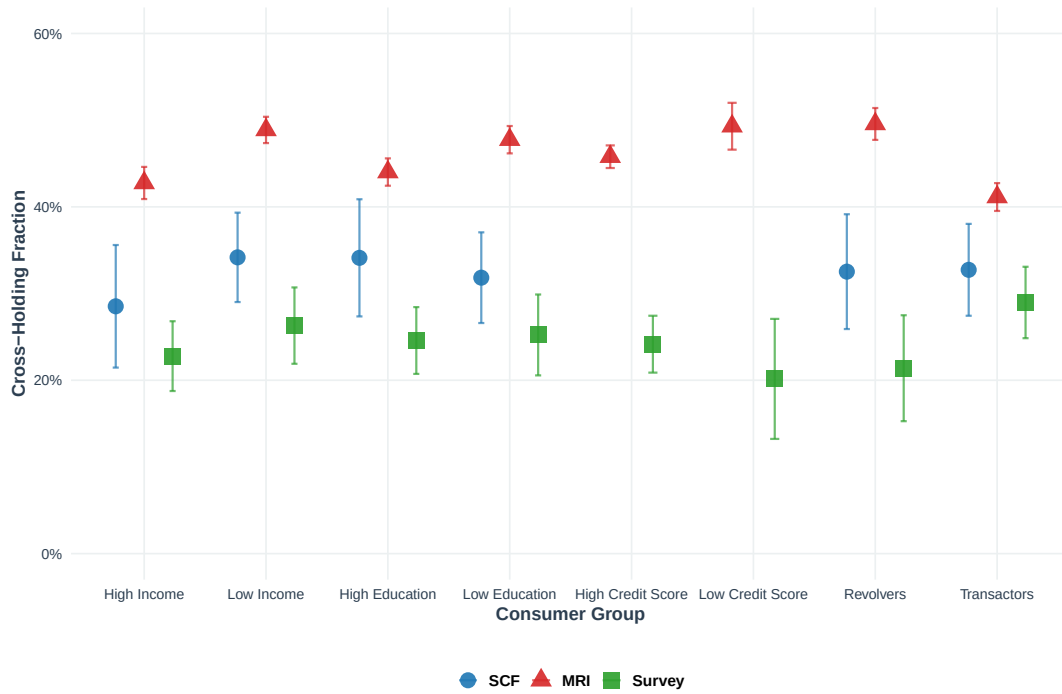
Notes: This figure shows the prevalence of cross-holding across banks. The top panel plots, for each bank, the ratio of credit card market shares among consumers who do and do not have a checking account at the bank. The bottom panel plots the analogous ratio for checking account market shares conditional on credit card ownership. Banks are sorted in descending order by their aggregate market shares in checking accounts.

Figure A.3: Sizes of Consideration Sets



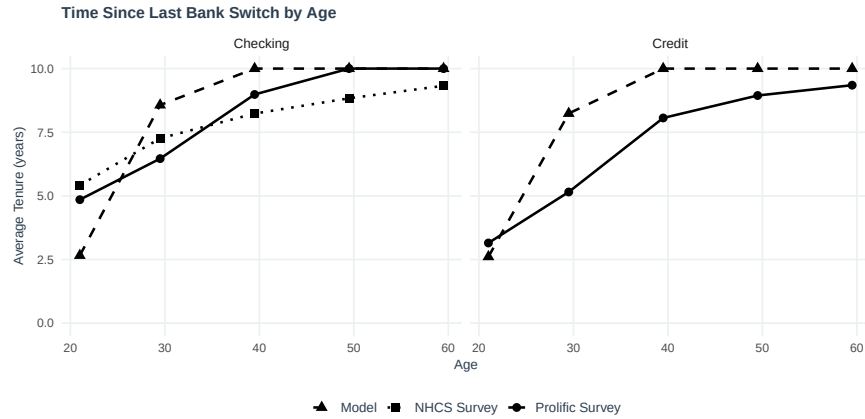
Notes: This figure shows the distributions of the sizes of the consideration sets for deposit accounts and credit cards in our Prolific Survey Sample. The consideration set includes each consumer’s bank of choice, and other banks they “looked into or reviewed” when choosing the product.

Figure A.4: Cross-holding by Consumer Subgroups: Detailed View



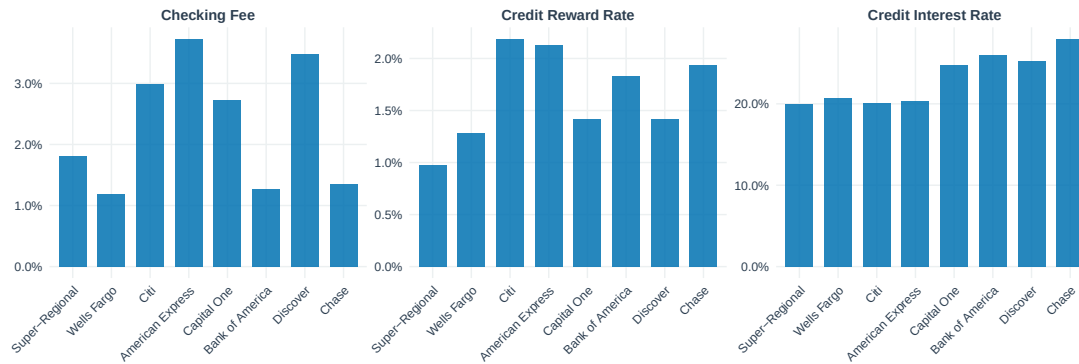
Notes: This figure provides a detailed breakdown of cross-holding rates by consumer subgroups. Each panel shows the rate of credit card cross-holding separately for different demographic characteristics: income, education, credit score, and credit card balance. Within each panel, we compare disadvantaged and advantaged groups as defined in the main text.

Figure A.5: Goodness of Fit: Tenure by Age



Notes: This figure compares model-predicted and empirical average tenure (time since the most recent bank switch) for checking accounts and credit cards, broken down by consumer age. Solid lines show Prolific survey averages; dashed lines show the model-implied analog from a forward simulation in which each synthetic consumer is assigned a first-account arrival age drawn uniformly from a fixed window and then experiences switching dynamics under the estimated CTMC until their current age.

Figure A.6: Observed Prices by Bank



Notes: This figure reports the observed prices used as the supply-side targets in the model: deposit interest rates, credit card reward rates, and credit card APRs by bank. Deposit rates are constructed from Y-14 quarterly bank holding company filings as the ratio of interest expense on domestic deposits to the total amount of domestic deposits. Reward expense per dollar of credit card spending is constructed from bank financial filings combined with payment volume data from the Nilson Report. APRs are from the CFPB Terms of Credit Card Plans (TCCP) survey.

Table A.1: Cross-holding and Consumer Characteristics

	1 Checking + 1 CC (1)	Main CC at Checking (1)	CC at Each Checking (1)
Age / 10	-0.10*** (0.01)	-0.10*** (0.01)	-0.12*** (0.01)
College Degree	0.00 (0.03)	0.02 (0.02)	0.04+ (0.02)
Log Income	-0.08*** (0.02)	-0.08*** (0.01)	-0.11*** (0.01)
CFO or Treasurer	-0.09 (0.15)	-0.32** (0.11)	-0.35* (0.14)
Online Banking	0.06+ (0.03)	0.03 (0.03)	-0.00 (0.03)
Invests in Stocks	-0.20*** (0.03)	-0.15*** (0.02)	-0.09*** (0.03)
Money Market	-0.09+ (0.05)	-0.08* (0.03)	-0.02 (0.04)
Tax-Advantaged Accounts	-0.19*** (0.03)	-0.15*** (0.02)	-0.20*** (0.02)
FICO > 780	-0.02 (0.04)	-0.03 (0.03)	0.01 (0.03)
Factor: Interest Rates	-0.13*** (0.03)	-0.11*** (0.02)	-0.08*** (0.02)
Factor: Reward Program	0.66*** (0.03)	0.61*** (0.02)	0.59*** (0.03)
Num.Obs.	28858	50100	50053
FE: num_cc^num_checking		X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes: This table shows results of logistic regressions estimating the relationships between consumer characteristics and the propensity to hold checking accounts and credit cards at the same bank. Column (1) restricts the sample to consumers with exactly one checking account and one credit card; the dependent variable equals one if both products are from the same bank. Columns (2) and (3) use the full sample of consumers with at least one checking account and one credit card. In Column (2), the dependent variable indicates whether the consumer has a checking account at the same bank as the primary credit card. In Column (3), the dependent variable indicates whether all pairs of checking accounts and credit cards are held at the same banks. Both columns flexibly control for the number of checking accounts and credit cards. *** implies significance at 0.001 level, ** 0.01, * 0.05, + 0.1.

B An Example of Consumer Demand Model

Consider a simple example with two banks and the outside option for each product. There are nine possible states:

$$x \in \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2)\}$$

Assume for simplicity that $\delta_{jk} = 0$ and $p_{jk} = 0$ for all products. Let $\chi = \log 2$, so that consumers are twice as likely to choose the credit card from the bank where they have the deposit account and vice versa. We can derive the Markov transition intensity matrix for the deposit account as:

$$Q_1 = \lambda_1 \times \begin{pmatrix} -2/3 & & 1/3 & & 1/3 & & & & \\ & -3/4 & & 1/2 & & 1/4 & & & \\ & & -3/4 & & 1/4 & & 1/2 & & \\ 1/3 & & & -2/3 & & 1/3 & & & \\ & 1/4 & & & -1/2 & & 1/4 & & \\ & & 1/4 & & & -3/4 & & 1/2 & \\ 1/3 & & & 1/3 & & & -2/3 & & \\ & 1/4 & & & 1/2 & & & -3/4 & \\ & & 1/4 & & & 1/4 & & & -1/2 \end{pmatrix}$$

This matrix illustrates key model features. The first row shows transitions from state $(0,0)$. Only 2 off-diagonal elements appear: consumers can move only to states holding a credit card at bank 0, since the matrix captures deposit-switching opportunities only. Given equal product qualities, the transition rate to each of those states is $\lambda_1/3$, so the total flow out of $(0,0)$ is $-2\lambda_1/3$.

The second row describes transitions out of the $(0,1)$ state. Here, the consumer already has a credit card at bank 1. With $\chi = \log 2$, they are twice as likely to choose the deposit account from the same bank. As a result, the transition rate to the $(1,1)$ state is $\lambda_1/2$, twice the transition rate to the $(2,1)$ state. The total flow out of the $(0,1)$ state then becomes $-3\lambda_1/4$. The third has a similar intuition as the second row, and the fourth row has a similar intuition as the first row.

The fifth row brings out another economic force at work. This row represents transitions from $(1,1)$, where the consumer is already cross-holding at bank 1. With $\chi_1 = \chi_2 = \log 2$, the consumer prefers keeping the deposit account at bank 1. Therefore, the transitions into $(0,1)$ and $(2,1)$ are reduced. This generates the empirical pattern that

consumers who cross-hold are less likely to consider switching and more likely to be at their bank for a longer period of time.

Similarly, the transition matrix for Q_2 is

$$Q_2 = \lambda_2 \times \begin{pmatrix} -2/3 & 1/3 & 1/3 & & & & & & \\ 1/3 & -2/3 & 1/3 & & & & & & \\ 1/3 & 1/3 & -2/3 & & & & & & \\ & & & -3/4 & 1/2 & 1/4 & & & \\ & & & 1/4 & -1/2 & 1/4 & & & \\ & & & 1/4 & 1/2 & -3/4 & & & \\ & & & & & & -3/4 & 1/4 & 1/2 \\ & & & & & & 1/4 & -3/4 & 1/2 \\ & & & & & & 1/4 & 1/4 & -1/2 \end{pmatrix}$$

and the transition intensity matrix for the entire system is $Q = Q_1 + Q_2$.

Now suppose $\lambda_1 = 0.1$ and $\lambda_2 = 0.5$. This reflects the idea that people switch deposit accounts rarely but switch credit cards quickly. The full transition matrix then looks like

$$Q = \begin{pmatrix} -0.4 & 0.17 & 0.17 & 0.03 & & & & 0.03 & \\ 0.17 & -0.41 & 0.17 & & 0.05 & & & & 0.02 \\ 0.17 & 0.17 & -0.41 & & & 0.02 & & & 0.05 \\ 0.03 & & & -0.44 & 0.25 & 0.12 & 0.03 & & \\ & 0.02 & & 0.12 & -0.3 & 0.12 & & 0.02 & \\ & & 0.02 & 0.12 & 0.25 & -0.45 & & & 0.05 \\ 0.03 & & & 0.03 & & & -0.44 & 0.12 & 0.25 \\ & 0.02 & & & 0.05 & & 0.12 & -0.45 & 0.25 \\ & & 0.02 & & & 0.02 & 0.12 & 0.12 & -0.3 \end{pmatrix}$$

Given this process, an approximate steady-state vector of market shares $\mathbf{s}$ is a left eigenvector of the matrix Q with eigenvalue 0, normalized to have a sum of 1:

$$\mathbf{s} = (0.09 \ 0.09 \ 0.09 \ 0.09 \ 0.18 \ 0.09 \ 0.09 \ 0.09 \ 0.18)$$

At the steady state, consumers are more likely to cross-hold both products at the same bank.

The matrix exponential for one time period is $S(1)$ and equals

$$\exp(Q) = \begin{pmatrix} 0.69 & 0.12 & 0.12 & 0.02 & 0.01 & 0.0 & 0.02 & 0.0 & 0.01 \\ 0.12 & 0.69 & 0.12 & 0.0 & 0.04 & 0.0 & 0.0 & 0.02 & 0.01 \\ 0.12 & 0.12 & 0.69 & 0.0 & 0.01 & 0.02 & 0.0 & 0.0 & 0.04 \\ 0.02 & 0.0 & 0.0 & 0.66 & 0.19 & 0.09 & 0.02 & 0.0 & 0.01 \\ 0.0 & 0.02 & 0.0 & 0.09 & 0.76 & 0.09 & 0.0 & 0.02 & 0.0 \\ 0.0 & 0.0 & 0.02 & 0.09 & 0.19 & 0.65 & 0.0 & 0.0 & 0.04 \\ 0.02 & 0.0 & 0.0 & 0.02 & 0.01 & 0.0 & 0.66 & 0.09 & 0.19 \\ 0.0 & 0.02 & 0.0 & 0.0 & 0.04 & 0.0 & 0.09 & 0.65 & 0.19 \\ 0.0 & 0.0 & 0.02 & 0.0 & 0.0 & 0.02 & 0.09 & 0.09 & 0.76 \end{pmatrix}$$

Note that all entries of the matrix exponential are strictly positive (entries displayed as 0.0 reflect rounding to one decimal place), even though the transition intensity matrix Q has many zeros. In the model, it is possible to transition from, say, $(1,1)$ to $(2,2)$ in a short time window if you receive both a deposit account and credit card switching shock in the time window. The matrix exponential encodes this possibility through the process of multiplying the matrix.

C Data and Balance Parameter Calibration

C.1 Balance Parameters in the Model

The supply-side first-order conditions require balance parameters b_k measuring the typical dollar amounts in checking deposits (b_1), credit card outstanding balances (b_2^{balance}), and annual credit card spending (b_2^{spend}). Banks' profit calculations depend on these parameters: for checking, revenue comes from the net interest margin (the difference between rates earned on deposits and rates paid to depositors); for credit cards, from interest on balances and interchange fees on transaction volume.

C.2 Data Sources and Calibration

We calibrate each balance parameter as the simple average of median values from independent survey sources, separately by consumer type (revolvers, who carry a credit card balance, and transactors, who pay in full each month). Three surveys contribute to the calibration. The Federal Reserve's 2022 Survey of Consumer Finances (SCF) provides household-level medians for all three parameters: deposits, credit card outstanding balances, and credit card spending. The National Household Credit Survey (NHCS) provides weighted median deposit balances for a nationally representative sample of 16,489 consumers. The MRI-Simmons USA Study provides median annual credit card spending derived from survey spending buckets across approximately 100,000 consumers. For each parameter, we average across all sources that measure it: deposit balances average the SCF and NHCS medians, credit card spending averages the SCF and MRI medians, and credit card outstanding balances use the SCF median alone (the only source that reports this variable by consumer type). Table C.2 reports the source-specific median values side by side.

The resulting calibrated values are as follows. For revolvers, the averaged values yield deposits of \$4,620, credit card outstanding balances of \$3,430, and credit card spending of \$3,936 per year. For transactors, the averaged values yield deposits of \$22,759 and credit card spending of \$14,295 per year, with zero outstanding balances by definition. The 4.9-fold difference in deposits between transactors and revolvers reflects meaningful heterogeneity in the financial profiles of these consumer groups: transactors tend to be more affluent and hold substantially higher deposit balances.

Multi-Source Comparison Table C.2 compares the source-specific median values across the SCF, MRI/NHCS, and two additional reference sources (Prolific and Nilson Report). The SCF and NHCS deposit medians are broadly consistent: \$3,740 versus \$5,500 for

revolvers and \$23,520 versus \$21,998 for transactors. The SCF and MRI credit card spending medians agree closely for revolvers (\$3,816 versus \$4,056) but diverge more for transactors (\$18,384 versus \$10,206), reflecting differences in survey design and spending measurement. Averaging across sources mitigates idiosyncratic measurement error in any single survey.

Table C.2: Balance Parameters: Multi-Source Comparison (SCF, MRI/NHCS, Prolific, Nilson)

Parameter	SCF	MRI/NHCS	Prolific	Nilson
<i>Panel A: Revolvers (Balance Carriers)</i>				
Deposits (\$)	3,740	5,500	1,100	—
CC Outstanding (\$)	3,430	—	880	6,657*
CC Spending (\$/yr)	3,816	4,056	2,820	\$0*
<i>Panel B: Transactors (Non-Carriers)</i>				
Deposits (\$)	23,520	21,998	9,000	—
CC Outstanding (\$)	\$0	—	—	\$0*
CC Spending (\$/yr)	18,384	10,206	8,400	25,602*

Notes: SCF values are survey medians from the Survey of Consumer Finances. MRI/NHCS: deposit medians from the National Household Credit Survey (NHCS); CC spending medians from MRI Simmons; neither source covers CC outstanding. Prolific values are survey medians from the Prolific Academic panel. *Nilson Report (2021–2022) values are allocated under the strong assumption that revolvers carry 100% of CC outstanding and transactors account for 100% of CC spending; Nilson does not include deposit data. Population shares: 43.3% revolvers, 56.7% transactors.

Table C.3: Checking Deposit Balances from NHCS: Mean versus Median

Consumer Type	Mean Balance	Median Balance	Ratio (Mean/Median)
Transactors	\$87,860	\$21,998	3.99
Revolvers	\$32,769	\$5,500	5.96
Pooled	\$61,297	\$10,999	5.57

Notes: This table compares mean and median checking deposit balances from the NHCS for transactors (consumers who pay credit card balances in full) and revolvers (consumers who carry balances). The pooled estimate combines both groups. The large mean-to-median ratios reflect the highly right-skewed distribution of deposit balances.

Rationale for Using Medians Deposit and credit card balances are highly right-skewed (mean-to-median ratios of 4 to 6; Table C.3), so medians better reflect the typical account holder’s balance than an average inflated by outliers. In unreported results, using means

leaves the qualitative cross-selling findings unchanged, though implied marginal cost levels shift as expected.

D Likelihood Construction Details

The main text decomposes the Prolific cardholder likelihood into three blocks: a static block $L^{\text{stat}} = D_1 \cdot D_2$ that captures the consideration set and bank choice at each historical decision moment (Equation 13), a stated-preference block L^{hyp} for the hypothetical shocks (Equation 15), and a dynamic block $L^{\text{dyn}} = \lambda_{i1} \cdot M \cdot \lambda_{i2} \cdot F$ that adds the timing layer (Equation 14). For brevity, parts of this appendix collect the static and dynamic blocks under the shorthand $L^s \equiv L^{\text{stat}} \cdot L^{\text{dyn}}$ since both depend on the timing parameterization (t_1, t_2) of the past switching history. This appendix gives the closed-form expressions for D_k , M , and F , enumerates the four first-account configurations and explains why specific terms drop in each, derives the closed-form integrals for censored switch times, formalizes the non-cardholder contribution, and lays out the regime-conditional formulas for the hypothetical-shock likelihood.

D.1 Closed-Form Expressions for the Static and Dynamic Terms

For any intensity matrix A , let $\tilde{A}$ denote the modified matrix with all off-diagonal elements zeroed out (i.e., no switches), and let $[A]_{x,y}$ denote the matrix element corresponding to a transition from state x to state y . We use these to write transitions that respect what we know happened during a given interval.

Static choice term D_k . The static block multiplies two switch-event probabilities $D_1 \cdot D_2$, where for each switch k ,

$$D_k = P(x_k^* \mid C_k, x(t_k^-), \theta_i) \cdot P(C_k \mid x(t_k^-), \theta_i), \quad (\text{D.1})$$

C_k is the observed consideration set, x_k^* is the observed bank choice, $P(x_k^* \mid C_k, x(t_k^-), \theta_i)$ is the logit choice probability conditional on the consideration set and the pre-switch state, and $P(C_k \mid x(t_k^-), \theta_i)$ is the probability of drawing that consideration set.

Intervening tenure M . The intervening tenure M is the matrix-exponential probability of the consumer's state evolution during the interval (t_2, t_1) between the two observed switches. Because the consumer's deposit account is fixed to the post-deposit-switch bank during this interval (no further deposit switches have occurred since t_1), we suppress deposit transitions in the generator while retaining the full credit-card

dynamics:

$$M = \left[\exp \left((\tilde{Q}_1 + Q_2) (t_1 - t_2) \right) \right]_{x(t_1^+), x(t_2^-)} . \quad (\text{D.2})$$

The matrix element corresponds to the transition from the post-deposit-switch state $x(t_1^+)$ to the pre-credit-card-switch state $x(t_2^-)$.

Final stability F . The final stability F is the matrix-exponential probability of remaining in the consumer's currently observed state x^* from the most recent switch through the survey date. Because both observed switches have already been accounted for, we suppress both products' transitions:

$$F = \left[\exp \left((\tilde{Q}_1 + \tilde{Q}_2) t_2 \right) \right]_{x^*, x^*} . \quad (\text{D.3})$$

The diagonal element gives the probability of remaining in x^* without any additional switches.

D.2 First-Account Cases

The likelihood in Equation 14 applies when neither the deposit nor the credit card is the consumer's first account of that type. When one or both accounts are first-time adoptions, certain terms drop from the likelihood. We focus throughout on the case where the consumer has held her current deposit account longer than her current credit card ($t_1 > t_2$); the symmetric case follows by relabeling products. Table D.4 summarizes the four configurations, and we explain each below.

Table D.4: Likelihood Terms by First-Account Status

Case	Deposit	CC	λ_1	D_1	M	λ_2	D_2	F
1	Not First	Not First	✓	✓	✓	✓	✓	✓
2	First	Not First		✓	✓	✓	✓	✓
3	Not First	First	✓	✓			✓	✓
4	First	First		✓			✓	✓

Two types of terms can drop from the likelihood, each for a distinct economic reason:

First, arrival rates drop for first accounts. The arrival rate λ_k governs the Poisson process of *reconsideration* opportunities, the rate at which an existing account holder reconsiders their current product. As discussed in the main text, λ_k conflates genuine inattention with the fixed costs of switching, and both frictions contribute to low observed switching rates. For first-time adoption, the relevant frictions may be quite different: the fixed costs of opening a new account may be much smaller than those of switching an

existing one, so consumers may consider acquiring a new product far more frequently than λ_k would imply. Since λ_k conflates these frictions, we condition on the adoption having occurred and retain only the choice component D_k , which captures bank selection and consideration set formation at the time of adoption. In Case 2, λ_1 drops because the deposit is a first adoption; in Case 3, λ_2 drops because the credit card is a first adoption; in Case 4, both drop.

Second, the intervening tenure term drops when the more recently switched product is first. The term M (Equation D.2) is the probability the consumer's state follows the specified dynamics between the deposit switch at t_1 and the credit card switch at t_2 . During this interval, the consumer holds the deposit account at the chosen bank while potentially switching credit cards under the generator $\tilde{Q}_1 + Q_2$. This term is informative only when the consumer actually held a credit card throughout the intervening period. When the credit card is a first account (Cases 3 and 4), the consumer did not hold any credit card during $[t_1, t_2]$, so the credit card state is the outside option ($x_2 = 0$) throughout with certainty. Because the consumer's state path during the intervening period is trivially determined, M contributes no additional information and drops from the likelihood.

Third, the final stability term and choice probabilities always appear. The final stability term F (Equation D.3) captures the probability of no further switches between the most recent event and the survey date, and is always relevant regardless of whether accounts are first adoptions. The choice probabilities D_k , which capture which bank was selected and what consideration set was formed, are likewise always observed and always enter the likelihood.

D.3 Censored Switch Times

When one or both account switches occurred more than $T_{\max}$ years ago, the exact timing is censored and we must integrate over all possible switch times. Because the CTMC yields matrix exponential terms, these integrals admit closed-form solutions.

Deposit switch censored ($t_1 > T_{\max}$, $t_2 \leq T_{\max}$). When the deposit switch is censored but the credit card switch is observed, we integrate the likelihood over all possible deposit switch times beyond the lookback window:

$$L^s(t_1, t_2 | \theta) = \int_{T_{\max}}^{\infty} L(u, t_2 | \theta) du$$

The key step is integrating the matrix exponential in the intervening-tenure term M (Equation D.2) over the censored horizon. Using the identity

$$\int_{T_{\max}}^{\infty} \exp(A u) du = -A^{-1} \exp(A T_{\max})$$

for a matrix A with strictly negative real eigenvalues, M is replaced by:

$$M_{\text{dep}}^{\text{cens}} = - \left[(\tilde{Q}_1 + Q_2)^{-1} \exp((\tilde{Q}_1 + Q_2)(T_{\max} - t_2)) \right]_{x(t_1^+) \rightarrow x(t_2^-)}$$

The matrix inverse $(\tilde{Q}_1 + Q_2)^{-1}$ arises from integrating the matrix exponential over the semi-infinite censored horizon. All other likelihood terms, including which arrival-rate and choice terms appear, depend on first-account status exactly as in Table D.4.

Both switches censored ($t_1 > T_{\max}, t_2 > T_{\max}$). When both switches are censored, we perform a double integral over all possible switch times:

$$L^s(t_1 > T_{\max}, t_2 > T_{\max} | \theta) = \int_{T_{\max}}^{\infty} \int_r^{\infty} L(u, r | \theta) du dr$$

Now both M and F change. The intervening tenure becomes:

$$M_{\text{both}}^{\text{cens}} = - \left[(\tilde{Q}_1 + Q_2)^{-1} \right]_{x(t_1^+) \rightarrow x(t_2^-)}$$

This arises from $M_{\text{dep}}^{\text{cens}}$ when the upper integration limit for the credit card switch time also extends to infinity: the inner integral over the deposit switch time u , evaluated at its lower limit $u = r$ where $\exp(\tilde{Q}_1(u - r)) = I$, leaves only the matrix inverse from the outer integral over r . The final stability becomes:

$$F_{\text{both}}^{\text{cens}} = \left(-[\tilde{Q}_1 + \tilde{Q}_2]_{x^*, x^*} \right)^{-1} \exp \left([\tilde{Q}_1 + \tilde{Q}_2]_{x^*, x^*} \cdot T_{\max} \right)$$

which integrates the scalar sojourn probability over $[T_{\max}, \infty)$. As with the single-censored case, the first-account adaptations from Table D.4 apply unchanged: the arrival-rate and choice terms adapt based on which accounts are first accounts, while M and F are replaced by their censored counterparts.

Non-cardholders The stated-preference and dynamic blocks both require a credit card. The stated-preference module asks consumers to evaluate price and unbundling shocks at a bank where they currently hold both products, and the dynamic block needs a

credit-card switch event to time. Prolific respondents who do not currently hold a credit card therefore enter only through a simplified version of the static block: a single choice probability that the consumer holds the outside option on credit cards, conditional on her checking-account holding, marginalized over the unobserved credit-card consideration set. Concretely, the non-cardholder static contribution is

$$L^{\text{stat,nc}}(\theta_i) = \sum_{C \in 2^{\mathcal{J}}} P(x_2^* = 0 \mid C, x_1^*, \theta_i) P(C \mid x_1^*, \theta_i),$$

which disciplines how attractive the credit-card outside option must be relative to the inside options. Non-cardholders contribute neither L^{hyp} nor L^{dyn} , because the survey did not elicit hypothetical responses or switching histories from this group.

D.4 Hypothetical-Shock Likelihood

Section V.C introduces the stated-preference likelihood

$$L^{\text{hyp}}(\theta_i) = \sum_C P(V, W, R, A \mid C, \theta_i) P(C \mid x^*, \theta_i).$$

This appendix derives the regime-conditional formulas for the conditional probability $P(V, W, R, A \mid C, \theta_i)$.

We model each respondent as drawing a single set of Gumbel preference shocks $\{\varepsilon_{ijk}\}$ at the time of the question; she then reports her hypothetical choices across all four scenarios using that draw. Define the consumer's choice probabilities under four progressively-degraded utility states for the focal product: the baseline (s_j), the unbundling shock (s'_j , where the ideal bank V no longer offers the complementary product $-k$), the price shock (s''_j , where the price at V becomes less attractive by Δp), and the complete removal of the focal product at V (s'''_j). Each s_j is a logit choice probability over the latent consideration set C .

Whether the price shock or the unbundling shock dominates the consumer's utility loss at the ideal bank depends on the realized $(\alpha_{ik}, \chi_{ik}^u)$. In the **price-dominant** case ($\alpha_{ik}\Delta p > \chi_{ik}^u \mathbb{1}\{V = x_{-k}^*\}$), any consumer who would switch under the unbundling shock would also switch under the price shock. The conditional probability collapses to

a stack of utility-gap differences:

$$P(V, W, R, A \mid C, \theta_i) = \begin{cases} s_0 & V = 0, \\ s_A''' - s_A'' & V = R = W \neq A, \\ s_A'' - s_A' & V = R \neq W = A, \\ s_A' - s_A & V \neq R = W = A, \\ 0 & \text{otherwise.} \end{cases}$$

Each non-zero case isolates the mass of consumers whose Gumbel draws place them in a particular utility window: just severe enough to trigger a switch under the stronger shock but not the weaker one.

In the **unbundling-dominant** case ($\alpha_{ik}\Delta p < \chi_{ik}^u \mathbb{1}\{V = x_{-k}^*\}$), the roles of W and R swap and the corresponding stack of differences uses the unbundling-side rather than price-side ordering. Because $(\alpha_{ik}, \chi_{ik}^u)$ are heterogeneous, the marginal likelihood mixes over the two regimes through the population distribution of types, which is what gives the model positive likelihood for every observed response pattern (including patterns that any single (α, χ^u) pair would assign zero probability to).

D.5 Model-Implied Tenure Distributions

We derive the model-implied distribution of account tenures from the estimated CTMC parameters. The key object is the *backward recurrence time*: the elapsed time since the consumer last switched to their current bank, as observed in a cross-sectional survey.

D.5.1 Phase-Type Backward Recurrence Time

Consider the tenure of a consumer's deposit account at bank j . Define $\mathcal{S}_1(j) = \{x : x_1 = j\}$ as the set of states in which the consumer holds a deposit account at bank j , varying over credit card banks $x_2 \in \{0, \dots, J\}$. While the consumer remains within $\mathcal{S}_1(j)$, they may switch credit cards but their deposit tenure continues to accumulate.

The sub-generator $Q_{\mathcal{S}} = Q[\mathcal{S}_1(j), \mathcal{S}_1(j)]$ governs the dynamics within $\mathcal{S}_1(j)$. Because $Q = Q_1 + Q_2$, this sub-generator captures both the (zero) within-set deposit transitions and the credit card transitions that occur while the consumer remains at deposit bank j . Exit from $\mathcal{S}_1(j)$, that is, switching the deposit account to a different bank, is encoded in the off-diagonal blocks of Q that are excluded from $Q_{\mathcal{S}}$. The sojourn time in $\mathcal{S}_1(j)$ therefore follows a phase-type distribution $\text{PH}(\alpha, Q_{\mathcal{S}})$, where α is the entry distribution described below.

D.5.2 Entry Distribution

The entry distribution α describes the composition of states within $\mathcal{S}_1(j)$ at the moment a consumer switches *into* deposit bank j . This distribution matters because a consumer's expected tenure at bank j depends on her credit card state x_2 at the time of entry. A consumer who enters as a cross-holder (j, j) faces lower exit rates because of the utility complementarity χ^u , and therefore tends to remain longer than a consumer who enters holding a credit card elsewhere. The entry rate into state $(j, x_2) \in \mathcal{S}_1(j)$ is:

$$\alpha_{x_2} = \sum_{j' \neq j} \bar{s}(j', x_2) \cdot q_1^{(j', x_2) \rightarrow (j, x_2)}$$

where $\bar{s}(j', x_2)$ is the steady-state probability of state (j', x_2) and $q_1^{(j', x_2) \rightarrow (j, x_2)}$ is the deposit transition rate from bank j' to j (holding credit card bank x_2 fixed). Normalizing so that $\sum_{x_2} \alpha_{x_2} = 1$ gives the probability distribution over credit card states at the moment of entry. Credit card tenure at bank j is defined analogously, with $\mathcal{S}_2(j) = \{x : x_2 = j\}$.

D.5.3 Backward Recurrence Distribution

In a stationary CTMC, the backward recurrence time B (the time elapsed since the most recent entry into $\mathcal{S}_k(j)$) has the following distribution:

$$f_B(t) = \frac{\alpha' \exp(Q_S t) \mathbf{1}}{\mu} \quad (\text{D.4})$$

$$P(B > t) = \frac{-\alpha' Q_S^{-1} \exp(Q_S t) \mathbf{1}}{\mu} \quad (\text{D.5})$$

where $\mathbf{1}$ is the vector of ones and $\mu = -\alpha' Q_S^{-1} \mathbf{1}$ is the mean sojourn time in $\mathcal{S}_k(j)$.

Intuitively, $\exp(Q_S t) \mathbf{1}$ gives the probability of remaining within $\mathcal{S}_k(j)$ for at least t years starting from each state. Averaging over the entry distribution α and normalizing by μ converts this into the backward recurrence density observed in a cross-section. The survival function in Equation (D.5) exploits the commutativity $Q_S^{-1} \exp(Q_S t) = \exp(Q_S t) Q_S^{-1}$ to precompute $Q_S^{-1} \mathbf{1}$ once.

D.5.4 Aggregation

We compute per-bank tenure distributions separately for each bank $j = 1, \dots, J$ and each product k . The overall tenure distribution for product k is the share-weighted average across banks:

$$f_B^{\text{overall}}(t) = \sum_{j=1}^J w_j f_B^{(j)}(t), \quad w_j = \frac{\bar{s}_j}{\sum_{j'=1}^J \bar{s}_{j'}}$$

where $\bar{s}_j$ is the marginal steady-state share of bank j for product k (excluding the outside option). We further combine the transactor and revolver subsamples using their sample frequency weights to produce a single predicted tenure distribution for comparison with the survey data. Because the demand parameters θ_i are heterogeneous, each type's tenure distribution is computed separately and then averaged over the Monte Carlo draws from the heterogeneity distribution.

D.6 Parametric Form of the Heterogeneity Distribution

We specify the bank-preference intercepts in levels and the structural primitives in logs, since α, χ^u, λ are positive on theoretical grounds. The heterogeneity distribution has block-diagonal structure:

1. **Bank preferences (levels).** For each bank j , the consumer's preference vector is jointly normal,

$$\begin{pmatrix} v_{ij1}^c \\ v_{ij2}^c \\ v_{ij1}^u \\ v_{ij2}^u \end{pmatrix} \sim \mathcal{N}(\bar{v}_j, \Sigma_v),$$

independent across consumers i and banks j , with Σ_v shared across banks. The off-diagonal entries of Σ_v allow consumers' preferences to correlate within a bank across products and across the consideration and utility margins, which is empirically central: we estimate the consideration–consideration correlation $\rho(v_1^c, v_2^c)$ to be close to one.

2. **Price sensitivity and utility complementarity (logs, per product).** For each product k , $(\log \alpha_{ik}, \log \chi_{ik}^u)$ is jointly normal,

$$\begin{pmatrix} \log \alpha_{ik} \\ \log \chi_{ik}^u \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_{\alpha,k} \\ \mu_{\chi^u,k} \end{pmatrix}, \Sigma_{(\alpha, \chi^u),k}\right),$$

with within-product correlation freely estimated. The two product- k blocks $(\log \alpha_{i1}, \log \chi_{i1}^u)$ and $(\log \alpha_{i2}, \log \chi_{i2}^u)$ are independent of each other. The independence reflects the survey design: each Prolific respondent is randomized into hypothetical-shock questions for only one of the two products, so the within-product joint distribution

is identified but the cross-product joint distribution is not.

3. **Arrival rates (logs).** The pair of log arrival rates is jointly normal,

$$\begin{pmatrix} \log \lambda_{i1} \\ \log \lambda_{i2} \end{pmatrix} \sim \mathcal{N}(\mu_\lambda, \Sigma_\lambda),$$

with their own 2×2 covariance. Tenure distributions for both products in the same consumer's life identify the cross-product correlation in arrival rates.

The consideration complementarity χ_k^c is treated as a homogeneous parameter, with population mean $\bar{\chi}_k^c$ identified by Fact 3 but no consumer-level variance. The consumer-level shifters η_{ik}^c and η_{ik}^u used in earlier specifications are not separately identified from ν^c and ν^u , so we drop them. We use log specifications for α, χ^u, λ because the model imposes positivity on these parameters; the bank-preference intercepts ν have no sign restriction and are modeled in levels.

D.7 MC-EM Algorithm and Implementation Details

This appendix gives the full pseudocode and the practical features of the implementation glossed over in the main text. Let X_i denote the data for observation i , $\phi(\cdot \mid \mu, \Sigma)$ the multivariate normal PDF with mean μ and covariance Σ , and $\ell = 1, 2, \dots, L$ the iteration index.

Algorithm.

1. **Initialize.** Set $\bar{\theta}^{(1)}$ and $\Sigma^{(1)}$ to starting values described in the initialization paragraph below, and set each observation's proposal distribution to the population prior, $\bar{\theta}_i^{(1)} = \bar{\theta}^{(1)}$ and $\Sigma_i^{(1)} = \Sigma^{(1)}$.
2. For iterations $\ell = 1, \dots, L$ or until convergence:
 - (a) **Draw.** For each consumer i , draw R proposals $\theta_{i,r}^{(\ell)} \sim \mathcal{N}(\bar{\theta}_i^{(\ell)}, \Sigma_i^{(\ell)})$. The proposal distribution is observation-specific and is updated each iteration based on the previous iteration's posterior summary for i .
 - (b) **Evaluate.** Compute $L(X_i \mid \theta_{i,r}^{(\ell)})$, the population prior density $p_r^{(\ell)} = \phi(\theta_{i,r}^{(\ell)} \mid \bar{\theta}^{(\ell)}, \Sigma^{(\ell)})$, and the proposal density $q_{i,r}^{(\ell)} = \phi(\theta_{i,r}^{(\ell)} \mid \bar{\theta}_i^{(\ell)}, \Sigma_i^{(\ell)})$.
 - (c) **E-step (importance weights).** Form normalized posterior weights

$$w_{i,r}^{(\ell)} = \frac{L(X_i \mid \theta_{i,r}^{(\ell)}) p_r^{(\ell)} / q_{i,r}^{(\ell)}}{\sum_{t=1}^R L(X_i \mid \theta_{i,t}^{(\ell)}) p_t^{(\ell)} / q_{i,t}^{(\ell)}}.$$

The ratio $p_r^{(\ell)} / q_{i,r}^{(\ell)}$ reweights each draw from the proposal back to the population prior; multiplying by the likelihood yields draws from the posterior $p(\theta_i \mid X_i, \bar{\theta}^{(\ell)}, \Sigma^{(\ell)})$ up to a normalizing constant.

- (d) **M-step (population parameters).** Update $\bar{\theta}^{(\ell+1)}$ and $\Sigma^{(\ell+1)}$ as weighted first and second moments of the simulated types:

$$\begin{aligned}\bar{\theta}^{(\ell+1)} &= \frac{1}{N} \sum_{i=1}^N \sum_{r=1}^R w_{i,r}^{(\ell)} \theta_{i,r}^{(\ell)}, \\ \Sigma^{(\ell+1)} &= \frac{1}{N} \sum_{i=1}^N \sum_{r=1}^R w_{i,r}^{(\ell)} (\theta_{i,r}^{(\ell)} - \bar{\theta}^{(\ell+1)}) (\theta_{i,r}^{(\ell)} - \bar{\theta}^{(\ell+1)})^\top.\end{aligned}$$

- (e) **Update proposal.** Refresh each consumer's proposal using her own posterior moments:

$$\begin{aligned}\bar{\theta}_i^{(\ell+1)} &= \sum_{r=1}^R w_{i,r}^{(\ell)} \theta_{i,r}^{(\ell)}, \\ \Sigma_i^{(\ell+1)} &= \sum_{r=1}^R w_{i,r}^{(\ell)} (\theta_{i,r}^{(\ell)} - \bar{\theta}_i^{(\ell+1)}) (\theta_{i,r}^{(\ell)} - \bar{\theta}_i^{(\ell+1)})^\top.\end{aligned}$$

Step 2(c) is the E-step implemented by importance sampling; step 2(d) is the M-step in closed form, since the weighted log-likelihood of the simulated types has a Gaussian-conjugate maximizer in $(\bar{\theta}, \Sigma_\theta)$. The algorithm departs from Train (2008) in step 2(a): rather than drawing all types from the population prior, we draw each consumer's types from her own posterior summary from the previous iteration, sharply reducing the variance of the importance weights and allowing the E-step to converge with a manageable number of draws per observation.

The remainder of this appendix describes practical features of the implementation needed to run reliably on samples of our size.

Phased estimation schedule. We estimate the model in four phases. The first phase fits only the static moments (the static-choice and hypothetical-choice likelihoods), holding the arrival rates λ_{ik} at an initial calibration and estimating only the preference and complementarity blocks. The second phase adds the dynamic past-switching likelihood with the preference and χ parameters frozen at their phase-one estimates, so that the first iteration can update $\bar{\lambda}$ and σ_λ in isolation. The third phase unfreezes preferences while keeping the complementarity block fixed. The fourth phase estimates all parameters jointly. The phased schedule avoids the instability that arises from simultaneously

updating parameters whose moments compete in the early iterations.

Adaptive proposal covariance. The proposal covariance $\Sigma_i^{(\ell)}$ in step 2(a) of the algorithm is a convex combination of the previous-iteration posterior covariance for observation i and the current-iteration population covariance, with the mixing weight adapted based on the effective sample size of the importance weights. When weights are concentrated on few draws (low ESS), we shrink the proposal toward the population covariance to broaden the search; when weights are close to uniform (high ESS), we rely more on the observation-specific posterior covariance. The mixing weight is also scaled by a diffusion factor greater than one to prevent the proposal from collapsing below the true posterior scale.

Draw regeneration. Each MC-EM iteration can either re-simulate draws from the current proposal or reuse the draws from the previous iteration with updated importance weights. We regenerate draws when any of three conditions holds: the effective sample size $(\sum_r w_{i,r}^{(\ell)})^2 / \sum_r (w_{i,r}^{(\ell)})^2$ drops below a per-consumer threshold, the population parameters $\bar{\theta}^{(\ell)}$ have drifted far enough from their values at the last regeneration, or a fixed number of iterations has elapsed since the last regeneration. Reusing draws across consecutive iterations is exact up to the importance ratio; regeneration is needed only when the proposal has moved far enough that the importance weights degenerate.

Block-diagonal covariance constraint. The heterogeneity covariance is block-diagonal across the four blocks defined in Appendix D.6: the bank-preference block Σ_v (4×4 , shared across banks); the per-product log blocks $\Sigma_{(\alpha, \chi^u), 1}$ and $\Sigma_{(\alpha, \chi^u), 2}$ (2×2 each, independent of each other); and the joint log arrival-rate block Σ_λ (2×2). Cross-block correlations are constrained to zero. After each M-step we project the updated covariance onto this block structure and enforce positive-definiteness by eigenvalue flooring.

Aggregate-share refinement. After the MC-EM iterations converge, we adjust the bank-specific utility intercepts $\bar{v}_{jk}^u$ to match aggregate market shares from the MRI-Simmons cross-section. Holding all other parameters fixed at their EM estimates, we iterate

$$\bar{v}_{jk}^{u,(\ell+1)} = \bar{v}_{jk}^{u,(\ell)} + \log s_{jk}^{\text{MRI}} - \log \hat{s}_{jk}(\bar{v}^{u,(\ell)}),$$

where $\hat{s}_{jk}(\bar{v}^u)$ is the model-implied steady-state share for bank-product cell (j, k) , averaged over the estimated type distribution by Monte Carlo. The contraction converges

quickly because the residual share gap is small after EM, and it leaves the heterogeneity, complementarity, and switching-rate estimates unchanged.

Initialization. We initialize $\bar{\theta}^{(1)}$ at moderate values consistent with the reduced-form moments. In particular, bank-preference intercepts match observed market shares and arrival rates match the average reciprocal tenure observed in the survey. $\Sigma^{(1)}$ is initialized as a block-diagonal matrix with moderate diagonal entries and identity correlation structure within blocks. Convergence is declared when the relative change in the population log-likelihood and in $\bar{\theta}$ falls below prespecified tolerances across five consecutive iterations.